Second-order distribution accuracy of explicit Euler on a fixed probability path

Dev D. Goyal

Abstract

In flow models, numerical solvers can approximate distributions more accurately than the individual trajectories used to generate them. We study when modifying the velocity field gives this improvement for explicit Euler without changing the exact distribution at any time. The freedom to modify the field in this way is known, and we use it to prove an error bound over a fixed time interval. Our sufficient condition requires motion along the material acceleration field to leave the current distribution unchanged at each fixed physical time. With smoothness, global Lipschitz, and moment assumptions, Euler has an error of $O(H^2)$ in the Wasserstein distance, where H is the largest step size. Gaussian and nonlinear radial examples show that second-order distribution accuracy is achievable, although particle error remains first order. The correction doubles the leading particle error coefficient for the Gaussian and increases it by a similar factor in the tested nonlinear example. On uniform grids with equal numbers of field evaluations, corrected Euler exactly matches base-field Heun in Gaussian distribution. It also gives distribution errors close to Heun's in the tested nonlinear example. A different Gaussian rotation gives Heun third-order distribution accuracy and second-order particle accuracy, so the field and solver must be considered together. The condition rules out variance-preserving interpolation between independent standard Gaussian noise and data with the same second moment, unless the distribution is constant. Correcting less structured paths remains an open problem.

1 Introduction

Flow models generate samples by moving random starting points through a velocity field (Lipman et al., 2023; Albergo et al., 2025). Numerical solvers approximate this motion by taking a sequence of steps. For a learned field, each velocity evaluation usually needs a neural-network pass, so fewer evaluations can reduce the cost of sampling. The aim of sampling is to reproduce a distribution, but this does not require every particle to follow its exact trajectory. We study when Euler can approximate the distribution more accurately than it approximates the individual trajectories.

We compare fields that give the same distribution at every time. We call this sequence of distributions the *probability path*. For example, particles in a circular Gaussian distribution can move directly outward or rotate as they move outward. The rotation leaves the angles uniform, so both fields give the same exact distributions. We call the outward field the *base field* and the field with added rotation the *corrected field*. Euler gives different approximations for these two fields, although their exact distributions agree.

In the expanding Gaussian example, the outward speed increases during an exact step, but Euler keeps the initial velocity. Its endpoint is therefore too close to the origin. Adding rotation gives the step a perpendicular displacement, which increases the radius and makes it more accurate. In Figure 1, the radius difference decreases from 0.222 to 0.113. The added rotation also increases the angle error, so the distance from the numerical particle to its exact endpoint increases from 0.222 to 0.439.

Our main result gives a condition under which this improvement continues over repeated Euler steps. The condition uses *material acceleration*, which is the change in velocity along an exact trajectory. In a step of length h, Euler leaves out a displacement whose leading term is $h^2/2$ times this acceleration. We require motion along the acceleration field to leave the current distribution unchanged when physical time is fixed.

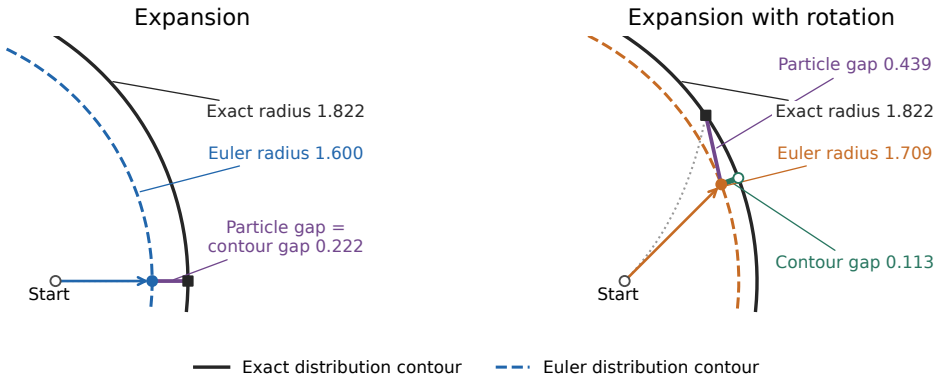


Figure 1: With added rotation, Euler gives a more accurate radius but a less accurate particle position. The purple line joins the Euler endpoint (filled circle) to the exact endpoint from the same starting point (black square). The green line gives the radius difference at a common angle, rather than the Wasserstein distance between the full distributions. The arcs show where the source unit circle moves for $a = 0.6$, $T = h = 1$, and $\Sigma_0 = I_2$. The exact radius is $e^{0.6} \approx 1.822$, while Euler gives 1.600 without rotation and $\sqrt{1.6^2 + 0.6^2} \approx 1.709$ with rotation. The corrected Euler angle is 20.6°, compared with the exact angle 34.4°.

Moving the starting points a small distance along this field then gives another set of starting points with the same distribution. Applying Euler to these shifted points supplies the missing displacement to leading order. The resulting points can be paired more closely with exact outputs from the original starts, giving a smaller bound on distribution error. This shift is part of the proof and does not change Euler's numerical update.

The modified-equation view explains the same cancellation (Hairer et al., 2006, Chapter IX). It compares Euler with the exact flow of the field $u_t - (h/2)\mathbf{a}_t$. With h fixed and physical time t evolving, this flow agrees with one Euler step up to $O(h^3)$, uniformly over compact sets of starting points. The acceleration condition leaves the continuity equation unchanged, so the modified field gives the same probability path under our global regularity assumptions. This expansion explains why the leading error disappears. We prove the error bound over the full time interval with a separate coupling argument, which does not require the full modified-equation series to converge (Appendix B).

Main result, informally. Under the acceleration condition and the smoothness, global Lipschitz, and moment assumptions of Theorem 3, Euler satisfies

$$\max_k W_2(\widehat{p}_k, p_{t_k}) = O(H^2).$$

Here H is the largest step size, p_{t_k} is the exact distribution, and $\widehat{p}_k$ is the Euler distribution. The Wasserstein distance W_2 is the smallest root mean squared distance over all ways of pairing the two distributions. The bound includes the errors from earlier steps and holds at every grid point in the fixed time interval. Particle error can still be first order, even when distribution error is second order.

The closest earlier result describes Euler's error in the expectation of a smooth test function (Hurault et al., 2026). The leading term depends on material acceleration and the exact motion during the remaining time. Our condition makes this term vanish after integration by parts. The remainder can still depend on the test function, so the expansion alone does not give a W_2 bound. Even matching several expectations need not make two distributions equal. For example, a variable that takes -1 and 1 with equal probability has the same mean and variance as a standard Gaussian. Work based on acceleration, strain, and vorticity instead compares the numerical and exact particles from the same starting point (Tao & Choi, 2026). Our distribution bound allows a different pairing.

The Gaussian example shows that distribution and particle error can have different orders. We then prove the distribution bound and give a nonlinear radial example with the same two orders. The leading error

coefficients are positive, so these examples achieve the stated rates. Further results show when the condition is locally necessary, which paths it rules out, and how the correction must change with the solver.

In both examples, the base field minimizes *kinetic action*, the time integral of average squared speed. Adding rotation increases this action but reduces Euler's distribution error. On uniform grids with equal numbers of function evaluations (NFE), corrected Euler gives exactly the same Gaussian distributions as base-field Heun. The field and solver therefore need to be considered together. This comparison does not give a correction for every path. Under our global assumptions, a gradient field can satisfy the acceleration condition only when almost every trajectory is straight and has constant velocity. Euler is already exact in that case.

Previous work shows how to change a velocity field without changing its probability path and how to minimize action on that path (Ambrosio et al., 2005; Liu, 2022; Pao-Huang et al., 2026). The Gaussian rotation family is also known (Lim, 2025). We use these results to prove the distribution error bound and show examples that achieve its order. Our other contributions are the local necessity result, the restrictions on probability paths, and the solver comparisons. Section 8 compares these contributions with earlier work.

2 Error measures and the acceleration condition

Term or symbol	Meaning	Definition
Probability path p_t	The distribution specified at each time t	Section 2.1
Local and global error	Error from one Euler step started from the exact distribution, and error accumulated through repeated steps	Section 2.1
Particle error	Root mean squared distance using the same starting point	Section 2.2
Coupling and W_2	A joint distribution with given marginals, and the least root mean squared distance over all such pairings	Eq. (2)
Material acceleration $\mathbf{a}_t$	Velocity change along an exact particle	Eq. (3)
Acceleration condition	$\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t = 0$ for every smooth compactly supported ϕ and every t	Definition 1
Kinetic action $\mathcal{A}_T$	Time integral of kinetic energy $\mathbb{E}_{p_t}\|u_t\|^2$	Eq. (5)
Second moment $M(t)$	$\mathbb{E}_{p_t}\|X\|^2$	Section 6

2.1 The probability path and Euler's method

A velocity field u generates the probability path $(p_t)_{0\le t\le T}$ when the solution of $\dot{X}_t = u(t, X_t)$, with $X_0 \sim p_0$, has distribution p_t at each time. Each p_t, called a *marginal distribution*, describes positions at one time without recording the particles' starting points. Different particle motions can therefore give the same probability path, as the outward and rotating Gaussian fields do. We assume finite second moments and write $\mathbb{E}$ for expectation, Law for distribution, and $\|\cdot\|$ for the Euclidean norm. Single bars $|\cdot|$ denote scalar absolute value or complex modulus, while $\|\cdot\|_{\mathrm{op}}$ denotes the induced operator norm of a linear or multilinear map.

On a grid $0 = t_0 < \cdots < t_N = T$, with step lengths $h_k = t_{k+1} - t_k$ and largest step $H = \max_k h_k$, Euler's method is

$$\widehat{X}_{k+1} = \widehat{X}_k + h_k u(t_k, \widehat{X}_k), \qquad \widehat{X}_0 \sim p_0, \qquad \widehat{p}_k = \mathrm{Law}(\widehat{X}_k). \tag{1}$$

For $u(x) = x$, a particle starting at one reaches 1.1 after an Euler step of length 0.1 and reaches 1.21 after the next step. The second step starts from the first numerical position, so it includes the error from the first step. We call the error after repeated steps the *global error*. The *local error* is the error from one step that starts from the exact distribution p_t. We compare fields on the same grid. When studying convergence, we decrease the step sizes while keeping both fields and the final time fixed.

2.2 Particle error and distribution error

A rotation shows why particle displacement and distribution change can differ. Rotating the uniform unit circle by 30° moves each point by $2\sin(15°) \approx 0.518$, although the distribution stays the same. We can pair each rotated point with an original point at the same location, giving zero distance between the distributions.

Such a pairing is a *coupling*, a joint distribution with the two given marginal distributions. The Wasserstein distance is the least root mean squared distance over these couplings,

$$W_2(\mu,\nu) = \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \int \|x-y\|^2 \,\mathrm{d}\gamma(x,y) \right)^{1/2}, \tag{2}$$

where $\Gamma(\mu,\nu)$ is the set of couplings of μ and ν. Particle error uses the numerical and exact particles from the same starting point. Its root mean squared value is $\|\widehat{X}_N - X_T\|_{L^2} = (\mathbb{E}\|\widehat{X}_N - X_T\|^2)^{1/2}$. This pairing is one of the couplings in Eq. (2), so distribution error cannot exceed particle error. Practical sampling metrics also compare distributions. For example, Fréchet Inception Distance uses the squared Wasserstein distance between Gaussian fits to feature distributions (Heusel et al., 2017). Our examples use W_2 between the full distributions.

We calculate the errors from Gaussian formulas or deterministic quadrature, so they have no sampling uncertainty. Smaller values mean better accuracy in all figures and tables. An $O(H^2)$ bound means that the error is at most a constant times H^2 for small H, but the error could decrease faster. A positive limit of error divided by h^2 shows that second order is the actual rate. In that case, halving a sufficiently small uniform step reduces the error by approximately a factor of four.

2.3 The acceleration condition

For a density p_t, the continuity equation $\partial_t p_t + \nabla\cdot(p_t u_t) = 0$ describes how the velocity field changes the distribution. Adding a field r_t with $\nabla\cdot(p_t r_t) = 0$ leaves this equation unchanged, although it can change how individual particles move. When the probability measure has no density, we express the same condition using expectations of smooth functions. We apply it to material acceleration and use *jerk* for the change in acceleration along an exact trajectory,

$$\mathbf{a}_t = \partial_t u_t + (D_x u_t)u_t, \qquad \mathbf{j}_t = \partial_t \mathbf{a}_t + (D_x \mathbf{a}_t)u_t. \tag{3}$$

Here $\mathbf{a}_t(X_t)$ and $\mathbf{j}_t(X_t)$ are the second and third time derivatives of the exact particle position. The term $(D_x u_t)u_t$ accounts for the change in velocity caused by the particle's motion through space.

Definition 1 (Distribution-preserving fields and the acceleration condition)**.** A field w has *zero weighted divergence under* a probability measure p if

$$\int \nabla\phi(x)\cdot w(x)\, p(\,\mathrm{d}x) = 0 \quad \text{for every } \phi \in C_c^\infty(\mathbb{R}^d), \tag{4}$$

where the tests are smooth and compactly supported. A velocity field u satisfies the *acceleration condition* along (p_t) if Eq. (4) holds with $w = \mathbf{a}_t$ and $p = p_t$ at every time.

Zero weighted divergence refers to the probability measure p, while ordinary zero divergence refers to volume. For example, rotation preserves a circularly symmetric distribution because it keeps the angle uniform. A constant rightward field also has zero ordinary divergence, but it moves a Gaussian to the right and changes its distribution. Under the regularity assumptions of Proposition 12, Eq. (4) means that the flow of w preserves p exactly. For acceleration, this means that motion along $\mathbf{a}_t$ preserves p_t while physical time t is held fixed.

2.4 Kinetic action as a comparison objective

Penalties on speed or velocity derivatives are used to make numerical integration easier (Finlay et al., 2020; Kelly et al., 2020; Lee et al., 2023). We compare Euler's distribution error with kinetic action, the time integral of average squared speed,

$$\mathcal{A}_T(u) = \int_0^T \mathbb{E}_{p_t}\|u(t,X)\|^2 \,\mathrm{d}t, \tag{5}$$

where the integrand is kinetic energy, defined here without the factor $1/2$. The base fields in Sections 3 and 5 minimize this action among finite-action fields that satisfy the same weak continuity equation. Their Euler distribution errors are nevertheless larger than those of the corrected fields. Section 6 proves this action comparison and shows which restrictions the acceleration condition places on the path.

3 Rotation separates the two error orders for a Gaussian

For a Gaussian path, we can find the correction by comparing how Euler and the exact flow change the radius. Let J be a quarter-turn in two dimensions, so Jx is perpendicular to x and has the same length. The field $ax + bJx$ combines expansion at rate $a > 0$ with rotation at rate b. An Euler step sends x to $(1 + ah)x + bhJx$ and multiplies the squared radius by $(1 + ah)^2 + b^2h^2$. The exact expansion gives $e^{2ah} = 1 + 2ah + 2a^2h^2 + O(h^3)$, while Euler without rotation misses one a^2h^2 term. Choosing $b = a$ supplies this term, and the same choice works as the step size changes.

For the probability path

$$p_t = \mathcal{N}(0, e^{2at} I_2), \qquad a > 0, \tag{6}$$

the base and corrected fields are therefore

$$v_0(x) = ax, \qquad v_1(x) = a(I + J)x. \tag{7}$$

These fields are part of the known Gaussian rotation family (Lim, 2025). We use them to compare action, distribution error, and particle error. A linear change of coordinates gives the same construction for every positive definite covariance in even dimension. Euler also reproduces an added affine mean exactly. Appendix A gives these extensions, and Figure 2 shows an elliptical Gaussian example.

Theorem 2 (Gaussian distribution and particle errors)**.** *For the fields in Eq. (7) on $[0, T]$ with $T > 0$, the base field v_0 minimizes kinetic action among finite-action fields satisfying the same weak continuity equation. The corrected field adds $a(e^{2aT} - 1) > 0$ to that action. Euler's numerical covariance is $q_{i,k} I_2$, where*

$$q_{0,k} = \prod_{j<k} (1 + ah_j)^2, \qquad q_{1,k} = \prod_{j<k} (1 + 2ah_j + 2a^2h_j^2). \tag{8}$$

At every grid point after the start,

$$0 < W_2(\widehat{p}_k^1, p_{t_k}) < W_2(\widehat{p}_k^0, p_{t_k}), \qquad W_2(\widehat{p}_k^i, p_{t_k}) = \sqrt{2}\,[e^{at_k} - \sqrt{q_{i,k}}]. \tag{9}$$

For uniform $h = T/N \to 0$, write $C_\Sigma = \sqrt{2} e^{aT}$. Then

$$W_2(\widehat{p}_N^0, p_T) = C_\Sigma \tfrac{1}{2} a^2 T h + O(h^2), \tag{10}$$

$$W_2(\widehat{p}_N^1, p_T) = C_\Sigma \tfrac{2}{3} a^3 T h^2 + O(h^3), \tag{11}$$

$$\|\widehat{X}_N^1 - X_T^1\|_{L^2} = C_\Sigma\, a^2 T h + O(h^2). \tag{12}$$

For $z = ah > 0$, the covariance factors satisfy $(1 + z)^2 < 1 + 2z + 2z^2 < e^{2z}$. Both numerical clouds are therefore too small, but the corrected cloud is closer to the exact size. Multiplying these factors keeps the same strict ordering on every increasing grid. To find the error orders, we take logarithms and add the errors over the steps. The one-step log-scale errors are $z - \log(1 + z) = z^2/2 + O(z^3)$ for the base field and $z - \frac{1}{2}\log(1 + 2z + 2z^2) = 2z^3/3 + O(z^4)$ for the corrected field. On a uniform grid, the sum over T/h steps gives first-order and second-order errors, respectively.

An optimal coupling pairs Gaussian points at the same angle, so the distribution error depends only on scale. Particle error also includes the angle error between particles from the same starting point. The base field has no angle error. Eqs. (10) and (12) show that adding the correction doubles the leading particle error coefficient, although distribution error becomes second order. The elliptical Gaussian in Figure 2 has the same behavior. Increasing the uniform step count from 32 to 64 reduces base distribution error from 1.51×10^{-2} to 7.63×10^{-3}, a factor of 1.98. Corrected distribution error decreases from 3.74×10^{-4} to 9.47×10^{-5}, a factor of 3.94. Appendix A gives the proof and bounds for unequal steps.

This improvement on every grid requires expansion. During contraction, the added rotation can make the numerical radius too large. For $a = -0.6$ and one step of length one, the base scale 0.4 is closer to the exact scale $e^{-0.6} \approx 0.549$ than the corrected scale $\sqrt{0.52} \approx 0.721$.

For the expanding circular Gaussian, the corrected acceleration is $2a^2Jx$. Motion along this field follows circles and preserves the distribution. We use the same property below to bound the error without assuming a Gaussian path.

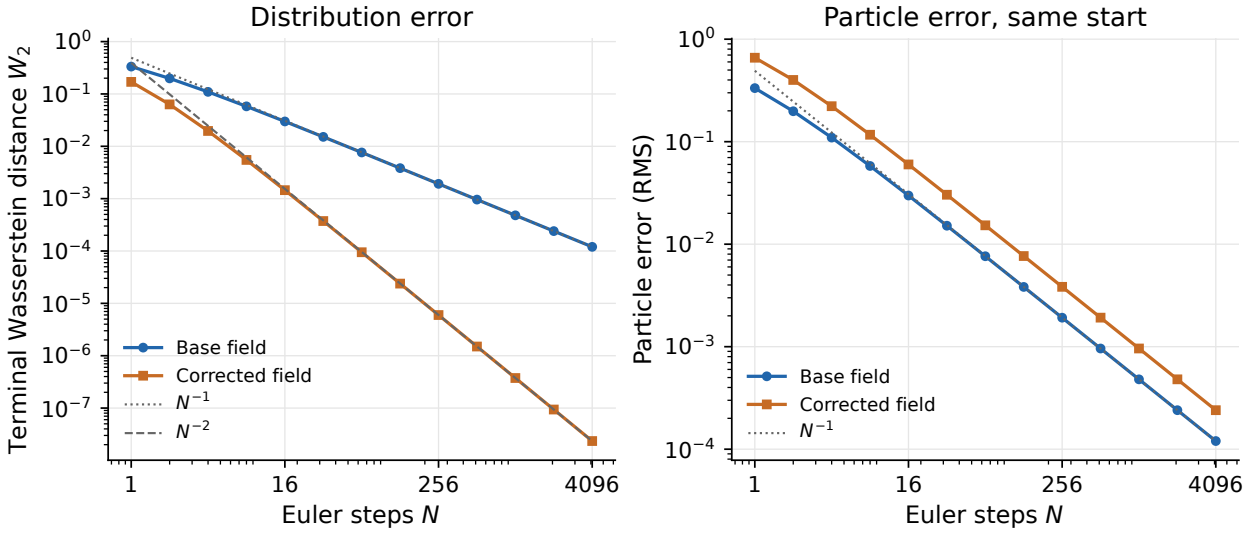


Figure 2: Adding rotation gives second-order Gaussian distribution error, although particle error remains first order with twice the base field's leading coefficient. The left panel shows W_2 between numerical and exact distributions. The right panel shows particle error from the same starting point. The fields stay fixed as the uniform step count N increases, and the reference lines show the N^{-1} and N^{-2} rates. Parameters are $a = 0.6$, $T = 1$, and $\Sigma_0 = \left(\begin{smallmatrix} 1.5 & 0.25 \\ 0.25 & 0.75 \end{smallmatrix}\right)$.

4 The acceleration condition bounds global distribution error

A velocity field can preserve a distribution even when its acceleration does not. For steady rotation $u(x) = Jx$ on the uniform unit circle, the velocity preserves the distribution, but the acceleration $\mathbf{a}(x) = J^2x = -x$ points inward. Motion along this acceleration shrinks the circle. The corrected expanding Gaussian has tangential acceleration, so motion along its acceleration preserves the distribution. This is the property we use to bound Euler's distribution error.

4.1 A coupling that reduces the local error

Let $E_{t,h}(x) = x + hu_t(x)$ be the Euler map and $F_{t,t+h}$ the exact flow over the same interval. Hold physical time t fixed, and let Ψ_t^τ be the flow that follows $\mathbf{a}_t$ for duration τ. Under the acceleration condition and the regularity assumptions below, $X \sim p_t$ and $\Psi_t^\tau X$ have the same distribution. Applying Euler to either starting point therefore gives the same output distribution. We can use the shifted start for Euler and the original start for exact motion to form a closer pairing.

The choice $\tau = h^2/2$ supplies the leading displacement that Euler omits. We write $O_{L^2}(h^q)$ for a remainder whose root mean squared norm is at most Ch^q, with C independent of h as $h \downarrow 0$. The shifted start satisfies $\Psi_t^{h^2/2}X = X + h^2\mathbf{a}_t(X)/2 + O_{L^2}(h^4)$. Evaluating velocity at this point changes the Euler displacement by only $O_{L^2}(h^3)$. Thus the root mean squared distance between $E_{t,h}(\Psi_t^{h^2/2}X)$ and $F_{t,t+h}(X)$ is $O(h^3)$, as Figure 3 shows. For the circular Gaussian with $a = 0.6$, the shift rotates the start by $2a^2\tau = 0.36h^2$.

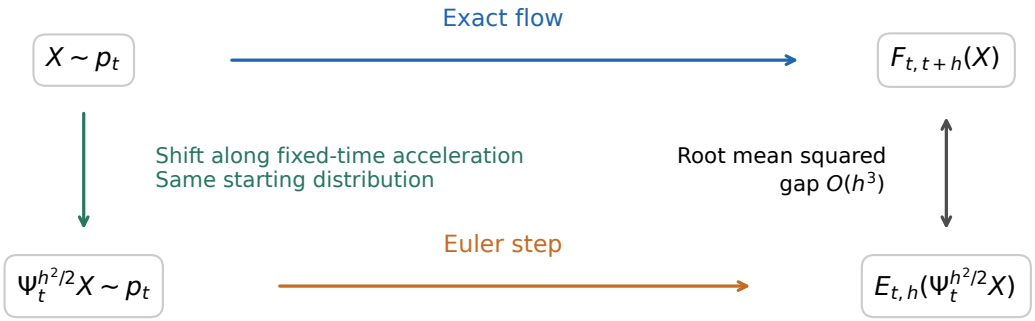


Figure 3: Shifting the start along the acceleration field gives a closer pairing and keeps Euler's output distribution unchanged. During the shift, physical time is held fixed and the distribution remains p_t. The shift supplies the leading displacement that Euler omits. We pair the Euler output from the shifted start with exact motion from the original start. Under Theorem 3, their root mean squared distance is $O(h^3)$.

To obtain the global bound, we also need to control how later steps increase this local error. Two continuous derivatives of velocity give the trajectory expansion. Global Lipschitz bounds control how far apart the outputs can be when the inputs differ, and mean square bounds allow us to integrate over the distribution. The theorem states these assumptions.

Theorem 3 (Global distribution error under the acceleration condition)**.** *Let p_0 have a finite second moment, and let p_t be the exact distributions of u. Suppose that u is C^2 on a neighborhood of $[0,T] \times \mathbb{R}^d$, globally Lipschitz in space with constant L, and $\sup_t \|u(t,0)\| < \infty$. Suppose that each $\mathbf{a}_t$ is globally Lipschitz with a common constant L_a, satisfies Eq. (4) under p_t, and has bounds*

$$C_a = \sup_t \|\mathbf{a}_t\|_{L^2(p_t)} < \infty, \qquad C_j = \sup_t \|\mathbf{j}_t\|_{L^2(p_t)} < \infty.$$

For Euler's method on every finite increasing grid,

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT} C_* \sum_j h_j^3 \le e^{LT} C_* T H^2, \qquad C_* = \frac{C_j}{6} + \frac{L C_a}{2} + \frac{T L_a C_a}{8}. \tag{13}$$

Each step adds an error bounded by a constant times h_j^3. The sum $\sum_j h_j^3 \le TH^2$ gives the total contribution of these errors, and e^{LT} bounds how much later Euler steps can increase them. The error is therefore at most a constant times H^2 at every grid point, without any Gaussian or symmetry assumption. The theorem gives a test for a particular field. It does not give a suitable field for every path, and Section 6 shows that some paths cannot satisfy the condition. Appendix B proves the bound using the local coupling and the Euler stability estimate $1 + Lh$.

4.2 Local necessity and weak error

The acceleration condition is also necessary if one Euler step from the exact distribution has error $o(h^2)$. Here $o(h^2)$ means that the error divided by h^2 tends to zero. For this rate to hold, the acceleration must preserve the distribution in the sense of Eq. (4).

Proposition 4 (Local converse)**.** *Let u and p_t satisfy the regularity, growth, and moment assumptions of Theorem 3, without imposing the acceleration condition. At a fixed $t < T$, suppose*

$$W_2((E_{t,h})_{\#} p_t, p_{t+h}) = o(h^2) \quad \text{as } h \downarrow 0.$$

Then $\mathbf{a}_t$ satisfies Eq. (4) under p_t.

Here $(E_{t,h})_{\#}p_t$ is the distribution obtained by applying Euler to a point drawn from p_t. The result concerns one step from the exact distribution. Errors across several steps can cancel at the final time, even when the acceleration condition fails. For example, the field $u(t,x) = \sin(2\pi t)e_1$ on $[0,1]$ with a Gaussian source has zero exact displacement and zero total Euler displacement on every uniform grid. Euler therefore gives the exact terminal distribution, although the acceleration changes the current distribution whenever it is nonzero. Appendix C gives the proof and this example.

For a smooth compactly supported test function ϕ, the difference between expectations after one Euler step and exact motion is

$$\mathbb{E}_{p_t}\phi(E_{t,h}X) - \mathbb{E}_{p_{t+h}}\phi = -\tfrac{1}{2}h^2 \int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t + o(h^2). \tag{14}$$

Under the proposition's assumptions, the absolute difference in expectations is at most the Lipschitz constant of ϕ times W_2. If $W_2 = o(h^2)$, the leading integral in Eq. (14) must therefore be zero. This gives the acceleration condition. Appendix C justifies the remainder, and Proposition 13 in Appendix C.1 gives an explicit cubic bound under a stronger moment assumption. Earlier global weak expansions follow expectation errors through the later exact flow (Talay & Tubaro, 1990; Hurault et al., 2026). The coupling in Theorem 3 bounds the distribution distance directly.

5 A nonlinear path has different error orders

The same difference in error orders is possible when the distribution changes shape. We use a radial distribution whose outer radii expand by larger relative factors. Unlike uniform expansion, this motion does not multiply all radii by the same number. Radial symmetry still gives an optimal pairing by radius, so we can compare errors when a covariance calculation is no longer enough.

We choose the rotation to cancel the radial part of acceleration. For the outward field $\alpha(r)x$, this part is $[\alpha(r)^2 + \alpha(r)r\alpha'(r)]x$. Adding the tangential velocity $\beta(r)Jx$ contributes $-\beta(r)^2 x$ to the radial acceleration. Choosing $\beta(r)^2 = \alpha(r)^2 + \alpha(r)r\alpha'(r)$ cancels the radial acceleration. The remaining tangential acceleration preserves a radial distribution. We choose an increasing α so that the relative expansion depends on radius. We also keep α bounded to control growth at large radii. With $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $a, \eta > 0$, define

$$\alpha(r) = a + \eta\frac{r^2}{1+r^2}, \qquad P(r) = \alpha(r)^2 + \alpha(r)r\alpha'(r), \qquad \beta(r) = \sqrt{P(r)}. \tag{15}$$

The positive function P gives a real rotation rate, and the resulting base and corrected fields are

$$u_0(x) = \alpha(\|x\|)x, \qquad u_1(x) = \alpha(\|x\|)x + \beta(\|x\|)Jx. \tag{16}$$

For both fields, the radius satisfies $\dot{R} = \alpha(R)R$. We write $R(t,r)$ for its solution from radius r, with $R(0,r) = r$. When the source is rotationally invariant, the added rotation keeps the angle uniform at each positive radius and leaves the origin fixed. The two fields therefore give the same distribution at every time, although their particles follow different trajectories. Since α increases with radius, the distribution changes shape whenever the source support contains two distinct positive radii.

For example, $a = 0.6$ and $\eta = 0.3$ give relative expansion rates $\alpha(1) = 0.75$ and $\alpha(2) = 0.84$. The outer circle therefore grows faster relative to its own radius. Figure 4 shows this shape change by dividing radius quantiles by the median radius. These ratios change along the nonlinear path. They would stay constant if every radius grew by the same factor.

The corrected field also satisfies the acceleration condition. Differentiation gives

$$\mathbf{a}_{u_1}(x) = \beta_a(\|x\|)Jx, \qquad \beta_a(r) = 2\alpha(r)\beta(r) + \alpha(r)r\beta'(r) \geq 2a^2. \tag{17}$$

This acceleration is tangent to each circle, so motion along it at fixed physical time preserves every radial distribution. The acceleration is nonzero away from the origin, so these trajectories are not straight with constant velocity as in Proposition 8. The next theorem shows that the correction gives smaller distribution

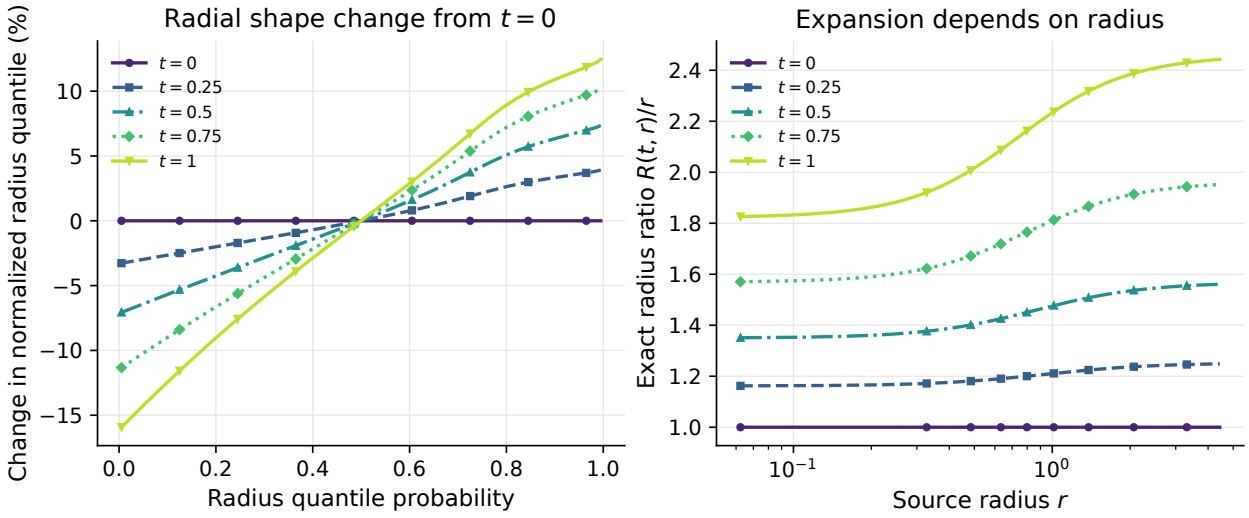


Figure 4: The nonlinear distribution changes shape because outer radii expand by larger relative factors. The left panel shows the percentage change, from $t = 0$, in each radius quantile divided by the median radius. Uniform scaling would leave these ratios unchanged. The right panel shows the expansion factor as a function of starting radius, with the parameters of Figure 5.

error on every grid. It also gives the exact error orders on uniform grids. Appendix D contains the regularity checks and proofs.

Theorem 5 (Nonlinear error separation on every grid)**.** *Let $a, \eta, T > 0$, and let p_0 be rotationally invariant on $\mathbb{R}^2$, with $0 < \mathbb{E}\|X_0\|^2 < \infty$. The fields in Eq. (16) are smooth and globally Lipschitz. They generate the same probability path. The base field minimizes kinetic action among finite-action fields satisfying the same weak continuity equation, and the correction strictly increases it. On every finite increasing time grid,*

$$0 < W_2(\widehat{p}_k^1, p_{t_k}) < W_2(\widehat{p}_k^0, p_{t_k}) \qquad (k \geq 1). \tag{18}$$

On uniform grids $h = T/N \to 0$, there are finite, strictly positive constants $\kappa_0, \kappa_1, \kappa_{\mathrm{p}}$ such that

$$\frac{W_2(\widehat{p}_N^0, p_T)}{h} \to \kappa_0, \qquad \frac{W_2(\widehat{p}_N^1, p_T)}{h^2} \to \kappa_1, \qquad \frac{\|\widehat{X}_N^1 - X_T^1\|_{L^2}}{h} \to \kappa_{\mathrm{p}}. \tag{19}$$

The error comparison follows from the one-step radius maps

$$G_{0,h}(r) = r(1 + h\alpha(r)), \qquad G_{1,h}(r) = r\sqrt{(1 + h\alpha(r))^2 + h^2 P(r)}.$$

For every $h, r > 0$, both maps increase with the starting radius and satisfy $G_{0,h}(r) < G_{1,h}(r) < R(h, r)$. The corrected radius is below the quadratic Taylor approximation of the exact radius. That approximation is below the exact radius because the exact radius has a positive third time derivative. Since the maps are increasing, the same ordering holds after repeated steps. Let $R_{i,k}(r)$ be the numerical radius after k steps under u_i. An optimal coupling pairs points at the same angle and matches their radii in increasing order. Since all radius maps are increasing, this pairs the radii that came from the same source radius. Thus

$$W_2(\widehat{p}_k^i, p_{t_k})^2 = \mathbb{E}\Big[\big(R_{i,k}(R_0) - R(t_k, R_0)\big)^2\Big], \qquad R_0 = \|X_0\|. \tag{20}$$

Equation (20) shows how the radius ordering gives the stated distribution error ordering. In one step, the radius falls short by positive leading terms of orders h^2 and h^3 for the base and corrected fields. After repeated steps, these give positive leading terms of orders h and h^2. The error bounds are proportional to the source radius, so a finite second moment is enough to average the squared errors.

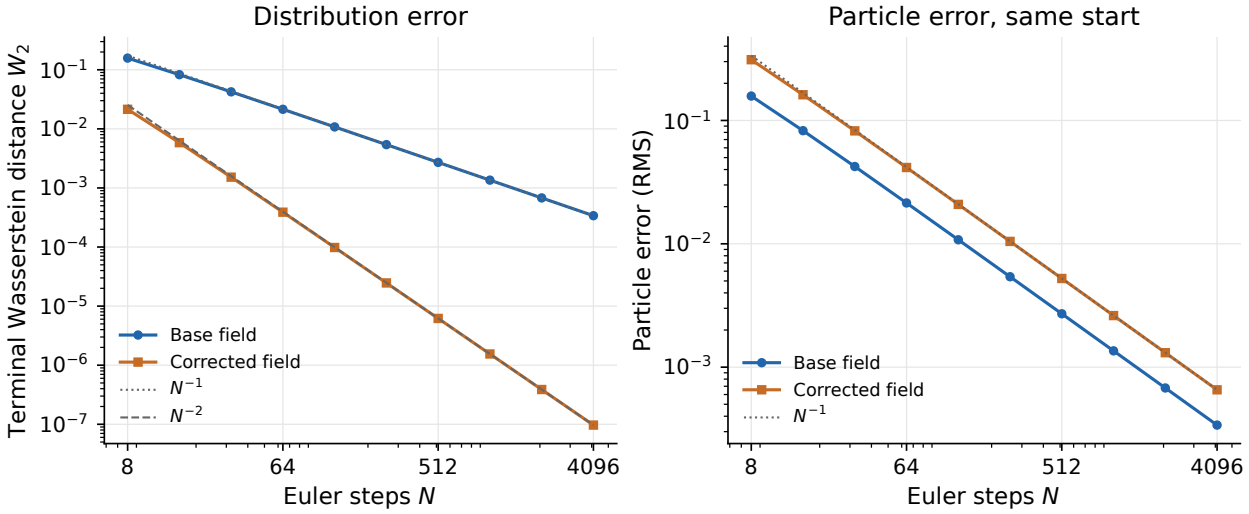


Figure 5: The corrected nonlinear field gives second-order distribution error, although particle error remains first order with a larger leading coefficient. The left panel uses Eq. (20) to measure distribution error, while the right panel compares particles from the same start. Both fields remain fixed as the uniform step size decreases, and the reference slopes give the rates in Eq. (19). The source is $0.6\mathcal{N}(0, 0.5^2 I_2) + 0.4\mathcal{N}(0, 1.5^2 I_2)$, with $a = 0.6$, $\eta = 0.3$, and $T = 1$.

Particle error remains first order because each corrected Euler step rotates too little by a term of order h^2. Over repeated steps, this gives a first-order angle error that is larger than the second-order radius error for sufficiently small steps. The base field keeps each angle fixed, so its particle and distribution errors are equal. For the mixture in Figure 5, the correction increases the leading particle error coefficient from 1.39 to 2.68, a factor of about 1.93. Appendix D derives these coefficients.

A fixed invertible affine transformation keeps the same error orders, and independent products give examples in higher dimensions. These extensions need not keep the same exact distances or strict error comparison between fields. Appendix D.5 states which properties hold.

Numerical verification. For the mixture in Figure 5, increasing the step count from 2,048 to 4,096 gives observed orders 0.9997, 1.9995, and 0.9998 for base distribution, corrected distribution, and corrected particle error, respectively. These values agree with the limiting orders in the theorem. We check numerical accuracy using Cartesian updates, an independent reference calculation, and finer quadrature. Appendix L and Table 4 give the results. The supplement includes the summary outputs and reproduction commands.

6 Which probability paths can satisfy the condition?

6.1 Mean, second moment, and kinetic action

The acceleration condition rules out some probability paths before we choose a velocity field. To see why, let $M(t) = \mathbb{E}\|X_t\|^2$ be the second moment. Its second derivative equals twice the kinetic energy plus $2\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)]$. Under the assumptions below, the acceleration condition makes the last term zero. The path therefore determines the kinetic energy of every field that satisfies the condition.

Proposition 6 (Second-moment restriction)**.** *Suppose exact trajectories and their first two time derivatives are continuous, and*

$$\mathbb{E} \sup_{0 \le t \le T} \left[\|X_t\|^2 + \|u_t(X_t)\|^2 + \|\mathbf{a}_t(X_t)\|^2 \right] < \infty.$$

If $\mathbf{a}_t$ *satisfies Eq. (4), then*

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} \mathbb{E}\|X_t\|^2 = 2\mathbb{E}\|u_t(X_t)\|^2 \ge 2K_{\min}(t), \tag{21}$$

where $K_{\min}(t)$ is the infimum of kinetic energy among fields satisfying the weak continuity equation at time t for the given probability path.

When the second moment is constant, Eq. (21) forces velocity to be zero almost everywhere. This rules out a changing distribution with constant second moment. For example, a centered noncircular Gaussian can rotate without changing its second moment. No field that generates this path can satisfy both the acceleration condition and the proposition's assumptions. A circular Gaussian stays unchanged under rotation, so its path does satisfy the condition with the zero field.

The condition also requires an affine mean, which moves at constant velocity. To prove this, we apply smooth cutoffs to the coordinate functions and let their supports expand. The acceleration condition gives zero second derivative of the mean. Integrating the second-moment identity also determines the action.

Corollary 7 (Affine mean and action fixed by the path)**.** *Under the assumptions and acceleration condition of Proposition 6, let $m(t) = \mathbb{E}X_t$, $M(t) = \mathbb{E}\|X_t\|^2$, and $\xi = \mathbb{E}u_0(X_0)$. Then*

$$m(t) = m(0) + t\xi, \qquad \mathcal{A}_T(u) = \tfrac{1}{2}\big[M'(T) - M'(0)\big]. \tag{22}$$

Thus any two fields that generate the same probability path and satisfy these assumptions have the same kinetic action.

The assumptions of Theorem 3 imply those of Proposition 6, as verified in Appendix G. The energy inequality also excludes any interval with $g'' < 0$ on a scaling path $p_t = \mathrm{Law}(g(t)X_0)$, provided $g > 0$ and $\mathbb{E}\|X_0\|^2 > 0$.

The action required by the acceleration condition need not be the minimum action on the path. Suppose a finite-action gradient field v^∇ generates the path and lies in the $L^2(p_t)$ closure of compactly supported smooth gradients for almost every t. Every finite-action correction r_t that preserves the weak continuity equation is orthogonal to this closure. The cross term in the squared velocity is therefore zero, which gives

$$\mathcal{A}_T(v^\nabla + r) = \mathcal{A}_T(v^\nabla) + \int_0^T \mathbb{E}_{p_t}\|r_t\|^2 \,\mathrm{d}t. \tag{23}$$

The added term is nonnegative. It is positive whenever the correction is nonzero in space-time L^2, so v^∇ minimizes action within this class (Ambrosio et al., 2005; Liu, 2022; Neklyudov et al., 2023). Here the distribution at every time is fixed. We are not minimizing over different paths with the same endpoints. Appendix H checks the closure assumption for the Gaussian and nonlinear examples.

6.2 Tests for generative probability paths

We next apply these moment restrictions to paths used in generative modeling. Let X_0 be standard Gaussian noise independent of data X_1 with finite second moment, and write $m_1 = \mathbb{E}X_1$ and $M_1 = \mathbb{E}\|X_1\|^2$. We test whether a field that generates each path can satisfy both the acceleration condition and the assumptions of Proposition 6.

Variance-preserving interpolation. Independence gives the following mean and second moment for $X_t = \rho_t X_1 + \sqrt{1-\rho_t^2}X_0$:

$$m(t) = \rho_t m_1, \qquad M(t) = d + (M_1 - d)\rho_t^2.$$

When $M_1 = d$, the second moment is constant, so only a stationary path can satisfy the stated condition and assumptions. This rules out a path from Gaussian noise to a different data distribution with the same second moment. When $M_1 \neq d$, the cosine schedule $\rho_t = \sin(\pi t/(2T))$ is also ruled out. In this case, $(\rho_t^2)''$ changes sign, so M'' is negative somewhere and contradicts Eq. (21). The schedule also has an affine mean only if $m_1 = 0$.

Independent linear interpolation. For $X_t = (1-t)X_0 + tX_1$ on $[0,1]$, each noise-data pair moves along a straight segment. The conditional-mean field averages the velocities of segments that pass through each

position. Its trajectories can therefore curve even though the original segments are straight. Conditional averaging cannot increase mean squared speed, so

$$\tfrac{1}{2}M''(t) = d + M_1 = \mathbb{E}\|X_1 - X_0\|^2 \geq \mathbb{E}\|\mathbb{E}[X_1 - X_0 \mid X_t]\|^2 \geq K_{\min}(t).$$

The last inequality uses $v_t(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$ as a candidate for $K_{\min}(t)$. This assumes enough regularity for the field to generate the path. With this assumption, independent linear interpolation passes the necessary moment test. Passing this test does not prove that a field satisfying the acceleration condition exists. Any such field must have constant kinetic energy $d + M_1$, and its action over the unit interval must also equal $d + M_1$.

6.3 Gradient fields force straight motion

The moment test does not decide whether a suitable field exists for independent linear interpolation. For global gradient fields, we can say more. Under all assumptions of the main theorem, the acceleration condition forces the acceleration to be zero.

Proposition 8 (Gradient fields have straight trajectories)**.** *Under all assumptions of Theorem 3, suppose each u_t is a global spatial gradient on $\mathbb{R}^d$. Then $\mathbf{a}_t = 0$ p_t-almost everywhere for every $t \in [0, T]$. For p_0-almost every start,*

$$X_t = X_0 + tu(0, X_0) \qquad (0 \leq t \leq T).$$

Euler is exact on every finite increasing grid for this same full-probability set of starts. Both particle error and distribution error are zero.

A gradient velocity has a symmetric Jacobian, which makes its material acceleration a gradient as well. The global assumptions put this acceleration in the weighted gradient closure. The acceleration condition makes it orthogonal to that closure, so the acceleration must be zero. Continuity along trajectories then gives straight motion for one full-probability set of starts. Appendix H.1 gives the proof.

For independent linear interpolation from standard Gaussian noise, the exact conditional-mean field is a gradient when the intermediate densities are smooth and positive (Lipman et al., 2023):

$$v_t(x) = \frac{x}{t} + \frac{1-t}{t}\nabla \log p_t(x), \qquad 0 < t < 1.$$

Both terms are spatial gradients. Under the theorem's global assumptions, the acceleration condition can therefore hold only when Euler is exact for almost every start. Exact probability-flow fields are also gradients when their forward drift is isotropic and linear (Song et al., 2021). A learned field need not equal these exact fields, so the assumptions must be checked for the learned field separately.

Regular displacement interpolation gives one class of straight trajectories. It pairs endpoints optimally and moves each pair at constant velocity. Appendix I states sufficient regularity conditions. Another class follows from Liu's straight rotated coupling between standard Gaussians (Liu, 2022, Example 3.5). We scale the target and use the quarter-turn pairing $X_1 = \sigma J X_0$. In even dimension, the resulting straight segments have the same distribution at every time as independent interpolation to $\mathcal{N}(0, \sigma^2 I)$. Appendix J proves this identity.

7 Solver choice and imperfect corrections

7.1 The useful correction depends on the solver

A correction selected for Euler need not improve another solver. We compare Euler with Heun's method, which averages the initial velocity with the velocity at an Euler prediction:

$$y = x + hu(t, x), \qquad x^+ = x + \tfrac{h}{2}\{u(t, x) + u(t + h, y)\}.$$

Heun uses two field evaluations per step and is also used in generative sampling (Karras et al., 2022). For the circular Gaussian field $(aI + bJ)x$, its one-step complex multiplier is $1 + h(a + ib) + \frac{1}{2}h^2(a + ib)^2$. The

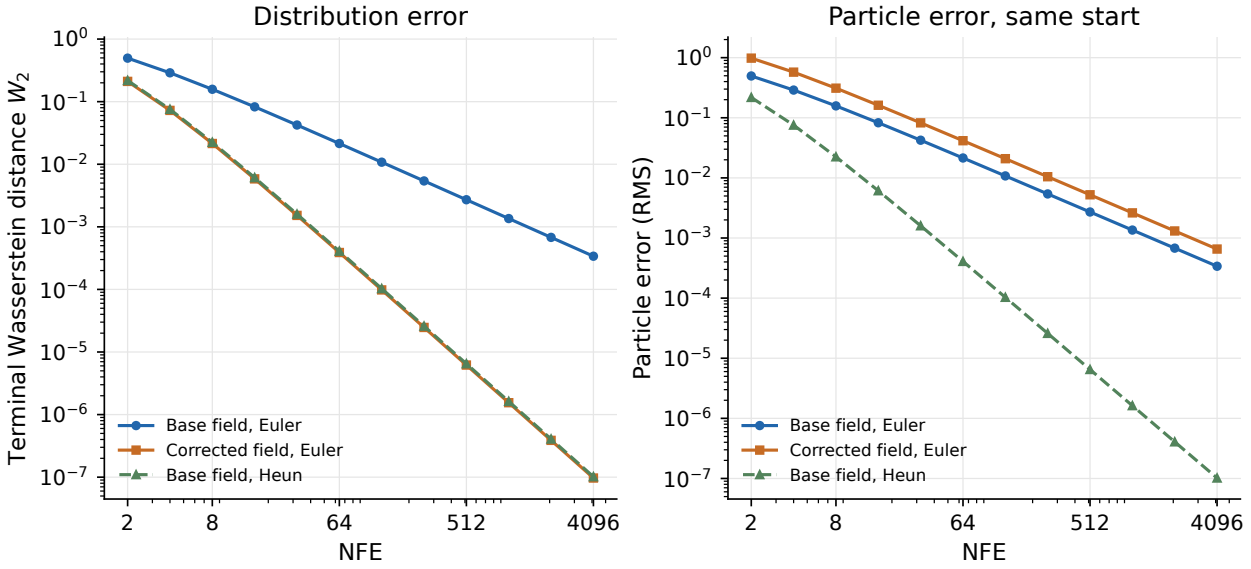


Figure 6: Corrected Euler closely matches base-field Heun in nonlinear distribution error at equal NFE, although it has larger particle error. The source and parameters are those of Figure 5, and each even N gives N uniform Euler steps or $N/2$ Heun steps of twice the length. The left and right panels compare distribution and particle errors, respectively. Base-field Heun has particle error equal to W_2.

magnitude of this multiplier gives the scale change, and its argument gives the rotation. Adding the log-scale errors over a uniform grid gives distribution error

$$C_\Sigma \frac{T|a^3 - 3ab^2|}{6} h^2 + O(h^3), \qquad C_\Sigma = \sqrt{2} e^{aT}. \tag{24}$$

The Euler choice $b = a$ doubles the coefficient in Eq. (24) compared with $b = 0$. Choosing $b = a/\sqrt{3}$ instead cancels this coefficient for Heun. Thus the useful correction depends on the solver. Appendix F derives the expansion and extends it to the Gaussian family of Appendix A.

Comparison at equal NFE. Heun uses twice as many evaluations per step as Euler. To compare them at equal NFE, we therefore use two corrected Euler steps of length h for each base-field Heun step of length $2h$. Both methods then give the same length multiplier,

$$\left(\sqrt{1 + 2ah + 2a^2h^2}\right)^2 = 1 + a(2h) + \tfrac{1}{2}a^2(2h)^2.$$

Proposition 9 (Gaussian equality at equal NFE)**.** *For the circular Gaussian fields in Theorem 2, let N be even and $h = T/N$. Compare corrected-field Euler with N steps of length h and base-field Heun with $N/2$ steps of length $2h$. Their distributions agree at every shared grid point. Each method uses N field evaluations over $[0, T]$.*

The identity extends to the Gaussian construction in Appendix A and to nonuniform grids with equal lengths within each consecutive pair of Euler steps. Appendix F.1 proves both statements.

The two methods have the same distributions at shared grid points, although their particle errors differ. Corrected Euler still has a first-order angle error, while base-field Heun has second-order particle error.

For the nonlinear mixture in Figure 6, corrected Euler has 3.3–4.7% smaller distribution error than base-field Heun over 2 to 4,096 NFE. Its particle error is much larger. Base-field Heun keeps each angle fixed and has an increasing radius map, so its particle error equals its distribution error. Table 1 gives the low-NFE values and ratios.

Table 1: At these low NFE values, corrected Euler has slightly smaller distribution error and larger particle error than base-field Heun. The setup is that of Figure 5, and the displayed W_2 equals particle error for both base-field methods. The final column gives the ratio of corrected Euler to base-field Heun distribution error, calculated from unrounded values.

NFE	Base field Euler W_2	Corrected field Euler W_2	Base field Heun W_2	Corrected field Euler particle error	Corrected Euler / base-field Heun W_2
2	4.96×10^{-1}	2.11×10^{-1}	2.18×10^{-1}	9.82×10^{-1}	0.9671
4	2.89×10^{-1}	7.25×10^{-2}	7.54×10^{-2}	5.74×10^{-1}	0.9609
8	1.58×10^{-1}	2.14×10^{-2}	2.24×10^{-2}	3.11×10^{-1}	0.9570
16	8.26×10^{-2}	5.84×10^{-3}	6.11×10^{-3}	1.62×10^{-1}	0.9548

With the rotation $b = a/\sqrt{3}$, Heun has third-order distribution error while particle error remains second order. We compare this corrected Heun method with Kutta's third-order Runge–Kutta method, denoted RK3, on the base field at equal NFE. Corrected Heun has 18.0–20.9% smaller distribution error at 12, 48, 192, and 384 NFE, although its particle error is larger. Appendix Table 3 gives the values. Appendix F.2 derives the coefficients, and Appendix Figure 8 shows convergence.

7.2 A residual acceleration adds a first-order term to the bound

An imperfect correction can keep the probability path unchanged without fully satisfying the acceleration condition. Write $\mathbf{a}_t = w_t + g_t$, where w_t satisfies Eq. (4) under p_t and g_t is the residual. Shifting the starting points along w_t supplies the part of Euler's missing displacement that preserves the distribution. The contribution from g_t remains.

Proposition 10 (Residual acceleration)**.** *Keep the assumptions on u and C_j from Theorem 3. Assume that w_t is C^1, globally Lipschitz with common constant L_w, and satisfies Eq. (4) under p_t. If $C_{w,2} = \sup_t \|w_t\|_{L^2(p_t)} < \infty$ and $\sup_t \|g_t\|_{L^2(p_t)} \le \varepsilon$, then*

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT} T \left(C_w H^2 + \tfrac{1}{2}\varepsilon H \right), \qquad C_w = \frac{C_j}{6} + \frac{L C_{w,2}}{2} + \frac{T L_w C_{w,2}}{8}. \tag{25}$$

In each step, the residual adds at most $\varepsilon h^2/2$ to the root mean squared distance between the paired points. Adding these contributions over the steps gives the εH term. If the other constants stay bounded, $\varepsilon = O(H)$ is enough for second-order accuracy. For example, halving both H and ε divides both terms in $H^2 + \varepsilon H$ by four. Halving only H divides the residual term by two. A fixed residual leaves a first-order term in the bound, but this does not mean that the actual error is always first order. Appendix E proves the bound and gives the sum over individual steps.

In the nonlinear example, a fixed mismatch in rotation gives first-order error. Replacing β by $\lambda\beta$, with $\lambda \ge 0$ fixed as the grid is refined, keeps the probability path unchanged but gives acceleration

$$\mathbf{a}_{u_\lambda}(x) = (1 - \lambda^2) P(r) x + \lambda \beta_a(r) J x.$$

For $\lambda \ne 1$, the remaining radial acceleration leads to $W_2(\widehat{p}_N^{\lambda}, p_T)/h \to |1 - \lambda^2| \kappa_0 > 0$, as proved in Appendix D.4. Even a small fixed mismatch therefore gives first-order convergence. It can still reduce error compared with the base field over a finite range of step sizes.

We can estimate where the mismatch starts to matter by equating its leading contribution $|1 - \lambda^2| \kappa_0 h$ with the exact correction's contribution $\kappa_1 h^2$. This gives

$$h_* \approx |1 - \lambda^2| \frac{\kappa_0}{\kappa_1}.$$

For the Gaussian path, the corresponding coefficients are $c_0^{\mathrm{G}} = C_\Sigma a^2 T/2$ and $c_1^{\mathrm{G}} = 2 C_\Sigma a^3 T/3$, where $C_\Sigma = e^{aT}\sqrt{\operatorname{tr} \Sigma_0}$. They give $h_*^{\mathrm{G}} \approx |1 - \lambda^2| c_0^{\mathrm{G}}/c_1^{\mathrm{G}} = 3|1 - \lambda^2|/(4a)$. At $a = 0.6$ on $[0, 1]$, the 2% mismatches

$\lambda = 0.98$ and 1.02 give about 20 steps, as shown in Appendix Figure 7. The first-order error dominates at sufficiently smaller step sizes. When $\lambda > 1$, the two contributions can have opposite signs and cancel near this estimate. Thus the estimate gives a range where both terms matter, rather than an exact threshold between orders.

8 Relation to previous work

Sampling error can change when we change the velocity field, the probability path, or the numerical method. Re-pairing endpoints can straighten trajectories, although it generally changes the intermediate distributions (Liu et al., 2023; Pooladian et al., 2023; Tong et al., 2024). Designing a new path can also reduce kinetic action or make numerical integration easier (Shaul et al., 2023). We keep the distribution at every time fixed. This gives us fewer choices of velocity field than in those path-design problems.

The freedom to add a suitable field with zero weighted divergence on a fixed probability path is already known (Ambrosio et al., 2005; Horvat & Pfister, 2024). Liu (2022, Sections 4–5) develops this freedom and the related minimization of quadratic action. The Gaussian rotation family appears in Lim (2025), and Flux Matching also treats fixed deterministic paths (Pao-Huang et al., 2026, Appendix E). We use these known constructions to study when Euler's distribution and particle errors have different orders.

The closest error expansion is the weak-error formula of Hurault et al. (2026, Eq. (4)). Its leading term describes how acceleration contributes to error after the later exact flow. The acceleration condition cancels this term after integration by parts. For the rotated fields here and other general drifts that satisfy Theorem 3, our coupling argument gives a direct finite-time W_2 bound. The nonlinear example has a positive leading coefficient, so second order is its actual distribution error rate. The local converse and path restrictions show where the condition is necessary and where it cannot hold.

Tao & Choi (2026) analyze particle error through acceleration components and state Euler exactness for straight displacement paths in Remark 7. They pair numerical and exact particles from the same starting point. In our rotating examples, another pairing removes the leading contribution to distribution error, although particle error from the same start remains first order.

Backward error analysis explains the same cancellation through Euler's first modified-field term $-(h/2)\mathbf{a}_t$. This term has zero weighted divergence under the acceleration condition (Hairer et al., 2006, Chapter IX). The formal expansion alone does not give our finite-time W_2 bound. Related rigorous results for stochastic systems include the truncated weak-operator expansions of Debussche & Faou (2012) and the invariant-measure order conditions of Abdulle et al. (2014). An invariant measure is a distribution that stays unchanged in time. Its averages can be approximated at a higher order than the finite-time weak order. Our result bounds W_2 for a changing distribution over a fixed interval. We also show examples with the stated error orders, a local converse, path restrictions, and solver comparisons.

Our regularity assumptions differ from those used to bound other errors in generative flows. Chen et al. (2023) assume globally Lipschitz true and estimated scores. For a probability-flow predictor with stochastic Langevin correctors, they obtain main endpoint bounds in total variation. Benton et al. (2024) allow a time-dependent spatial Lipschitz constant for the learned velocity and bound W_2 between two exact flows. We study the Euler discretization of an exact field. Our bound requires global Lipschitz bounds on both velocity and acceleration, together with the stated smoothness and moment assumptions. These conditions must be checked separately when applying the result to a learned field.

Gaussian distribution-error formulas are available from prior work (Pierret & Galerne, 2025). Our nonlinear example shows that the two error orders can also differ when covariance calculations are not enough. An optimal pairing of radii makes this comparison possible.

Bespoke solvers change the numerical method while keeping the field fixed (Shaul et al., 2024). We change the field while keeping the method and probability path fixed, then compare this with changing the solver at equal NFE. Exponential integrators solve a selected linear part exactly (Zhang & Chen, 2023; Lu et al., 2022). In our centered Gaussian example, a constant matrix defines the entire field, so its matrix exponential

gives exact motion. The diagonal Gaussian class of Claustre et al. (2026) does not include these rotating fields.

9 Discussion

Euler can approximate a distribution to second order while its particles remain only first-order accurate. Theorem 3 gives a sufficient acceleration condition and an explicit finite-time bound. The nonlinear example shows that both error orders are achievable. This matters for sampling because the target is a distribution. Applications that need trajectories from given starting points must also control particle error.

The useful correction depends on the numerical solver. The Gaussian rotation chosen for Euler doubles Heun's leading distribution error coefficient. A different rotation improves Heun's distribution error order. The field and solver must therefore be considered together.

At equal NFE on uniform grids, corrected Euler and base-field Heun give identical Gaussian distributions. Their distribution errors are also close in the tested nonlinear example, although corrected Euler has larger particle error. Equal evaluation counts do not mean equal runtime, since the cost of evaluating each field also matters.

Minimum kinetic action need not give the smallest numerical distribution error, even when the distribution at every intermediate time is fixed. In both examples, adding rotation increases action and reduces Euler's distribution error. The positive added action is specific to these constructions. It is not a universal lower bound. A field chosen separately for each fixed grid can have arbitrarily small added action if its time derivatives have no common bound, as Appendix K shows.

The acceleration condition determines action from the second moment and limits which paths can satisfy it. Under the global assumptions, a gradient field must give straight trajectories with constant velocity for almost every start. This also applies to the exact conditional-mean field of independent linear interpolation from standard Gaussian noise whenever those assumptions hold. The Gaussian, radial, affine, and product examples show that corrections are possible in their stated settings. The moment tests alone do not tell us how to construct a correction for another path.

The bound requires global regularity and moment assumptions, and its stability factor e^{LT} can be large. For approximate corrections, Proposition 10 adds a first-order residual term to the bound. This does not mean that every fixed residual gives first-order error. Applying the result to a learned field requires the stated acceleration decomposition, which a small divergence penalty alone need not give. The bound measures discretization error relative to the field's exact path. Error between a learned path and the intended path needs a separate estimate.

The probability path alone does not determine Euler's distribution accuracy. The bound shows how preserving the distribution under motion along acceleration improves the error estimate. The examples show that this improvement can come with larger particle errors. Constructing corrections for less structured paths remains open. For learned fields, we also need a usable bound on residual acceleration and a bound on probability-path error. Finite-time conditions for higher-order and stochastic solvers remain open questions.

AI assistance. Artificial intelligence tools assisted with literature checks, mathematical analysis, code, numerical verification, and manuscript preparation. The models used were GPT 6 Astra, GPT 6.1 Sol, and GPT 5.6 Sol in Codex, and Opus 5.5 in Claude Code. These tools checked mathematical arguments by separate derivations and proof review. They checked numerical results using closed-form calculations, direct particle updates, independent reference equations, and quadrature refinements. They also compared the closest prior-work claims with the cited source versions. The author independently checked all proofs, numerical results, and cited claims. The author remains responsible for the content of this manuscript.

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The appendices give the proofs and numerical details for the paper. In the Gaussian example, distribution error depends on scale, while particle error also depends on angle. The nonlinear example shows the same difference when the distribution changes shape. The general proof shows how the acceleration condition improves distribution accuracy in both examples. The table below gives the location of each result.

Main claim or comparison	Support
Finite-time W_2 bound and its coupling	Appendix B
Minimum action and the gradient-field consequence	Appendix H
Gaussian distribution and particle errors	Appendix A
Nonlinear rates, coefficient error, and extensions	Appendix D
Residual acceleration bound	Appendix E
Local converse and terminal cancellation	Appendix C
Solver dependence and equal NFE	Appendix F
Mean, second-moment, and action restrictions	Appendix G
Scaled rotated Gaussian coupling	Appendix J
Weak expansion and regular displacement interpolation	Appendices C.1, I
Arbitrarily small action cost on one grid	Appendix K
Population calculations and refinement results	Appendix L

For a map F, the pushforward $F_{\#}\mu$ is the distribution of $F(X)$ when $X \sim \mu$. We write $\mathcal{P}_2(\mathbb{R}^d)$ for probability measures with finite second moment, and define the root mean squared size of a vector field w by

$$\|w\|_{L^2(p)} = \left(\int \|w(x)\|^2 p(\,\mathrm{d}x) \right)^{1/2}.$$

For a random vector Z, $\|Z\|_{L^2}$ is its root mean squared norm. Table 2 lists the main symbols.

Table 2: Notation used in the proofs. Matrices and coefficients specific to an example are defined with that example.

Symbol	Meaning
p_t, $\widehat{p}_k$	Exact distribution at time t and Euler distribution at node k
u_t, r_t	Velocity field and a correction added to that field
$\mathbf{a}_t$, $\mathbf{j}_t$	Second and third time derivatives along an exact particle
T, h_k, H	Final time, step length, and largest step length
W_2	Distribution distance from the best root mean squared pairing
$\mathcal{A}_T$	Kinetic action over the time interval $[0, T]$
$F_{s,t}$	Exact particle flow from time s to time t
Ψ_t^τ	Auxiliary acceleration flow with physical time t held fixed
L, L_a	Common spatial Lipschitz bounds for velocity and acceleration
C_a, C_j	Uniform $L^2(p_t)$ bounds for acceleration and its material derivative
Σ_0, Ω	Initial Gaussian covariance and its weighted rotation matrix
C_0, K_t	Gaussian covariance root and skew matrix in the rotation representation
$C_{w,2}$, C_w	L^2 bound for w_t and constant in the residual bound
α, β, β_a	Nonlinear radial rate, rotation rate, and tangential acceleration coefficient
$\mathcal{C}_i$, Θ_1	Leading radius-error and angle-error coefficient functions
$\mathcal{T}$, G, d_{aff}	Affine map, its matrix, and its shift
$R(t,r)$, $R_{i,k}(r)$	Exact and numerical radii from starting radius r

A Fixed Gaussian correction and recursive errors

The Gaussian result also holds when the mean moves at constant velocity and the covariance is any positive definite matrix in even dimension. Assume $T, a > 0$ and $\Sigma_0 = C_0 C_0^\top$, with $J^\top = -J$, $J^\top J = I$, and

$\Omega = C_0 J C_0^{-1}$. For $m_t = m_0 + t\xi$, the fields $v_0(t,x) = \xi + a(x - m_t)$ and $v_1(t,x) = \xi + a(I + \Omega)(x - m_t)$ both generate $p_t = \mathcal{N}(m_t, e^{2at}\Sigma_0)$. These parameters stay fixed as the grid changes. The base field v_0 minimizes action among finite-action fields satisfying the same weak continuity equation. The correction increases action by

$$\mathcal{A}_T(v_1) - \mathcal{A}_T(v_0) = a^2 \operatorname{tr}(\Sigma_0) \frac{e^{2aT} - 1}{2a} > 0.$$

Euler reproduces m_t exactly and has covariance $q_{i,k}\Sigma_0$, where

$$q_{0,k} = \prod_{j<k} (1 + ah_j)^2, \qquad q_{1,k} = \prod_{j<k} (1 + 2ah_j + 2a^2 h_j^2).$$

Euler's distribution error is $\sqrt{\operatorname{tr}(\Sigma_0)}[e^{at_k} - \sqrt{q_{i,k}}]$. The corrected error is positive and strictly smaller for every $k \geq 1$. Equations (10)–(12) hold with $C_\Sigma = \sqrt{\operatorname{tr}(\Sigma_0)} e^{aT}$. Setting $m_0 = \xi = 0$ and $\Sigma_0 = I_2$ gives Theorem 2.

Write

$$I_a(T) = \int_0^T e^{2at}\, \mathrm{d}t = \frac{e^{2aT} - 1}{2a}.$$

Subtracting the mean and applying the inverse covariance factor gives a Gaussian source with identity covariance. These coordinates are called whitened coordinates. Each corrected Euler step changes scale and angle, but rotation leaves the Gaussian distribution unchanged. Distribution error therefore depends only on scale. Particles from the same starting point also have an angle error. The scale error has a positive second-order coefficient, while the angle error has a positive first-order coefficient.

The correction belongs to the Gaussian family of Lim (2025, Proposition 5.5). For a single Gaussian with $\Sigma_t = e^{2at} C_0 C_0^\top$, the skew matrix $K_t = b e^{-2at} C_0^{-\top} J C_0^{-1}$ gives $\Sigma_t K_t (x - m_t) = b\Omega(x - m_t)$. This form lets us calculate kinetic action and Euler error over the full time interval.

Proof of Theorem 2 and its Gaussian extension. **Common path and action.** The assumptions on J imply $J^2 = -I$. Direct matrix multiplication verifies

$$\Omega\Sigma_t + \Sigma_t \Omega^\top = 0, \qquad \Omega\Sigma_t\Omega^\top = \Sigma_t, \qquad \Sigma_t = e^{2at}\Sigma_0.$$

The centered flows $e^{at} I$ and $e^{at} C_0 e^{atJ} C_0^{-1}$ both give covariance Σ_t and mean m_t. Both flows are global and affine, so the distributions remain Gaussian. Equal means and covariances therefore give the same distribution at every intermediate time.

For $Y \sim \mathcal{N}(0, \Sigma_t)$, we have $\mathbb{E}\|\Omega Y\|^2 = \operatorname{tr}(\Sigma_t)$ and $\mathbb{E}[Y^\top \Omega Y] = \operatorname{tr}(\Omega\Sigma_t) = 0$, since $\Omega\Sigma_t$ is skew. The cross terms with ξ also have zero expectation because Y is centered. Expanding the squared speed and integrating over time therefore gives the action identities.

The base field is the gradient of $\xi^\top x + (a/2)\|x - m_t\|^2$. This potential has quadratic growth, so the projection argument in Appendix H applies. The base field uniquely minimizes action up to almost-everywhere equality. For the continuous fields in the class $\mathcal{V}(p)$ defined there, positive Gaussian density gives equality at every time and position. In particular,

$$\mathcal{A}_T(v_0) = T\|\xi\|^2 + a^2 \operatorname{tr}(\Sigma_0) I_a(T).$$

Recursive covariance and strict improvement. Let $r_{i,k} = \sqrt{q_{i,k}}$ and $\widehat{p}_k^{\,i} = \widehat{p}_k^{\,v_i}$ for $i = 0, 1$. Euler reproduces the affine mean exactly, while its centered whitened matrices are $(1 + ah_j)I$ and $(1 + ah_j)I + ah_j J$. The second matrix multiplied by its transpose is $(1 + 2ah_j + 2a^2 h_j^2)I$, so repeated covariance updates give the product formulas. For equal-mean Gaussians with covariances $r^2\Sigma_0$ and $s^2\Sigma_0$, W_2 equals $|r - s|\sqrt{\operatorname{tr}(\Sigma_0)}$.

For $z > 0$,

$$(1 + z)^2 < 1 + 2z + 2z^2 < e^{2z}.$$

The first difference is z^2, while the second inequality follows from the positive remaining terms in the exponential series. Taking square roots and multiplying over the steps gives $r_{0,k} < r_{1,k} < e^{at_k}$ for every $k \geq 1$. Both numerical distributions have too small a scale, but the corrected scale is closer to the exact value. The corrected error is strictly smaller at every grid point, so its maximum and sum are also smaller.

Accumulated error and convergence rates. To find the convergence orders, we calculate the scale error over all steps. Taking logarithms changes the product of step factors into a sum. The positive error terms for each step are

$$f_0(z) = z - \log(1+z), \qquad f_1(z) = z - \tfrac{1}{2}\log(1+2z+2z^2).$$

Their derivatives are

$$f_0'(z) = \frac{z}{1+z}, \qquad f_1'(z) = \frac{2z^2}{1+2z+2z^2}.$$

Integrate from zero to obtain $0 < f_0(z) \leq z^2/2$ and $0 < f_1(z) \leq 2z^3/3$. Put $\Delta_{i,k}^{\log} = \sum_{j<k} f_i(ah_j)$. Then $r_{i,k} = e^{at_k} e^{-\Delta_{i,k}^{\log}}$ and

$$W_2(\widehat{p}_k^i, p_{t_k}) = \sqrt{\operatorname{tr}(\Sigma_0)} e^{at_k}(1 - e^{-\Delta_{i,k}^{\log}}).$$

Use $1 - e^{-\Delta^{\log}} \leq \Delta^{\log}$, $\sum_{j<k} h_j^2 \leq t_k H$, and $\sum_{j<k} h_j^3 \leq t_k H^2$ to obtain the explicit unequal-step bounds

$$\max_k W_2(\widehat{p}_k^0, p_{t_k}) \leq \sqrt{\operatorname{tr}\,\Sigma_0} e^{aT} \frac{a^2 T}{2} H, \qquad \max_k W_2(\widehat{p}_k^1, p_{t_k}) \leq \sqrt{\operatorname{tr}\,\Sigma_0} e^{aT} \frac{2a^3 T}{3} H^2.$$

Near zero,

$$f_0(z) = \tfrac{1}{2}z^2 - \tfrac{1}{3}z^3 + O(z^4), \qquad f_1(z) = \tfrac{2}{3}z^3 - z^4 + O(z^5).$$

On the uniform grid,

$$\Delta_{0,N}^{\log} = \frac{a^2 T}{2} h + O(h^2), \qquad \Delta_{1,N}^{\log} = \frac{2a^3 T}{3} h^2 + O(h^3).$$

Substitute these expansions into $1 - e^{-\Delta^{\log}} = \Delta^{\log} + O((\Delta^{\log})^2)$ to obtain the terminal expansions. □

Proposition 11 (Particle error for the fixed correction)**.** *Pair the exact and numerical particles for v_1 through the same $X_0 \sim p_0$. On a uniform grid, set*

$$r_N = (1 + 2ah + 2a^2h^2)^{N/2}, \qquad \theta_N = N \arctan \frac{ah}{1+ah}.$$

Their terminal mean squared difference is

$$\mathbb{E}\|\widehat{X}_N - X_T\|^2 = \operatorname{tr}(\Sigma_0) \left[(r_N - e^{aT})^2 + 4 r_N e^{aT} \sin^2 \frac{\theta_N - aT}{2} \right].$$

Consequently, their root mean squared difference satisfies

$$\left(\mathbb{E}\|\widehat{X}_N - X_T\|^2 \right)^{1/2} = \sqrt{\operatorname{tr}(\Sigma_0)} e^{aT} a^2 T\, h + O(h^2).$$

Proof. Particle error compares the exact and numerical maps from the same starting vector, so it includes their angle difference. In whitened coordinates these maps are $e^{aT} e^{aTJ}$ and $r_N e^{\theta_N J}$. The difference multiplied by its transpose is a scalar matrix with factor $r_N^2 + e^{2aT} - 2 r_N e^{aT} \cos(\theta_N - aT)$. Multiplication by C_0 and $C_0^\top$ makes its trace equal to this factor times the initial covariance trace, proving the formula. The numerical mean is exact.

Expanding the numerical angle shows that

$$\theta_N - aT = -a^2 T h + O(h^2).$$

The radial scale difference is $O(h^2)$ by the theorem. The angle difference is first order and gives the leading particle error through the sine term. An optimal distribution pairing matches angles, so the angle difference does not contribute to W_2. □

B A distribution bound over a fixed time interval

We bound distribution error by moving Euler's starting point along the acceleration field while holding physical time fixed. If this motion preserves the starting distribution, Euler gives the same output distribution. Its output can then be paired more closely with exact motion from the original starting point. Zero weighted divergence ensures that the starting distribution stays exactly the same. A suitable shift supplies the leading displacement that Euler omits. Preserving selected averages, or preserving the distribution only to first order, would not be enough for this coupling.

Let $F_{s,t}$ be the exact flow of u, and let $p_t = (F_{0,t})_{\#}p_0$. Define

$$\mathbf{a}_t = \partial_t u_t + (D_x u_t)u_t, \qquad \mathbf{j}_t = \partial_t \mathbf{a}_t + (D_x \mathbf{a}_t)u_t.$$

These fields are the second and third time derivatives along an exact particle. A field w has *zero weighted divergence under* a measure p if

$$\int w \cdot \nabla\phi \, \mathrm{d}p = 0 \quad \text{for every } \phi \in C_c^\infty(\mathbb{R}^d). \tag{26}$$

The tests in C_c^∞ are smooth and vanish outside a bounded set. The definition therefore applies even when the measure has no density. If a density exists, the condition is the weak form of $\nabla\cdot(pw) = 0$. The auxiliary flow is autonomous, which means that its field stays fixed as auxiliary time changes.

Proposition 12 (Preservation with physical time fixed)**.** *Let p be a probability measure and let $w \in C^1(\mathbb{R}^d;\mathbb{R}^d)$ be globally Lipschitz. If w has zero weighted divergence under p, its autonomous flow Ψ^τ satisfies $(\Psi^\tau)_{\#}p = p$ for every $\tau \in \mathbb{R}$.*

Proof. Global Lipschitz continuity gives a flow at all positive and negative times, with Ψ^τ a C^1 diffeomorphism and inverse $\Psi^{-\tau}$. For a smooth compactly supported test ϕ, the support of $\phi\circ\Psi^\tau$ lies in the image of the support of ϕ under this continuous inverse. The composition therefore has compact support and is C^1, although it need not be infinitely differentiable. Convolution extends Eq. (26) to C_c^1 tests, so it applies to the composition.

The autonomous flow satisfies $D\Psi^\tau(x)w(x) = w(\Psi^\tau(x))$. For $\phi \in C_c^\infty$,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int \phi(\Psi^\tau(x))p(\,\mathrm{d}x) = \int w(x)\cdot\nabla(\phi\circ\Psi^\tau)(x)p(\,\mathrm{d}x) = 0.$$

On each bounded time interval, one compact set contains the supports of these integrands, where both the integrands and their derivatives are bounded. Differentiation under the integral is therefore valid. The derivative is zero, so every test expectation stays constant and the measures are equal. □

Because the auxiliary flow preserves the distribution exactly, we can form a coupling with local error bounded by a constant times h^3. Euler's Lipschitz bound then controls how much later steps increase this error. We restate Theorem 3 with the assumptions needed for these two estimates.

> **Theorem 3, restated.** Let $T > 0$ and $p_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Suppose u is C^2 on a neighborhood of $[0,T]\times\mathbb{R}^d$, globally Lipschitz in space with a common constant L, and $\sup_{t\in[0,T]}\|u(t,0)\| < \infty$. Suppose each $\mathbf{a}_t$ is globally Lipschitz with a common constant L_a and has zero weighted divergence under p_t. Assume
>
> $$C_a = \sup_{0\le t\le T}\|\mathbf{a}_t\|_{L^2(p_t)} < \infty, \qquad C_j = \sup_{0\le t\le T}\|\mathbf{j}_t\|_{L^2(p_t)} < \infty.$$
>
> For every finite increasing grid $0 = t_0 < \cdots < t_N = T$, put $h_k = t_{k+1} - t_k$, $H = \max_k h_k$, and $E_{t,h}(x) = x + hu(t,x)$. Euler's method means
>
> $$\widehat{p}_0 = p_0, \qquad \widehat{p}_{k+1} = (E_{t_k,h_k})_{\#}\widehat{p}_k.$$

Then

$$\begin{aligned}\max_{0\le k\le N} W_2(\widehat{p}_k, p_{t_k}) &\le e^{LT}\left(\frac{C_j}{6}+\frac{LC_a}{2}+\frac{TL_aC_a}{8}\right)\sum_{j=0}^{N-1} h_j^3 \\ &\le e^{LT}\left(\frac{C_j}{6}+\frac{LC_a}{2}+\frac{TL_aC_a}{8}\right)TH^2.\end{aligned} \tag{27}$$

Proof of Theorem 3. The growth and Lipschitz assumptions give an exact flow on the full interval with finite second moments. For $X \sim p_t$, Taylor's formula shows the displacement that Euler omits:

$$F_{t,t+h}(X) = X + hu_t(X) + \frac{h^2}{2}\mathbf{a}_t(X) + \rho_{t,h}.$$

The remainder integrates the third trajectory derivative, which has the stated L^2 bound because the intermediate distribution is p_{t+s}. Minkowski's inequality gives

$$\|\rho_{t,h}\|_{L^2} \le \frac{1}{2}\int_0^h (h-s)^2 \|\mathbf{j}_{t+s}(X_{t+s})\|_{L^2}\,\mathrm{d}s \le \frac{C_j}{6}h^3. \tag{28}$$

Let Ψ_t^τ be the flow of $\mathbf{a}_t$ with physical time t fixed. By Proposition 12, $\Psi_t^\tau X \sim p_t$ for every auxiliary time τ, so the bound C_a applies throughout the motion. Integrating acceleration bounds the displacement, while integrating its change from the initial value bounds the remainder. For $\tau \ge 0$,

$$\|\Psi_t^\tau X - X\|_{L^2} \le C_a\tau, \qquad \|\Psi_t^\tau X - X - \tau\mathbf{a}_t(X)\|_{L^2} \le \int_0^\tau L_aC_a v\,\mathrm{d}v = \frac{L_aC_a}{2}\tau^2.$$

Choose $\tau = h^2/2$. Euler's output from the shifted start still has distribution $(E_{t,h})_\# p_t$. Pairing this output with $F_{t,t+h}(X)$ gives the Wasserstein bound

$$\begin{aligned}\Psi_t^{h^2/2}X &= X + \tfrac{1}{2}h^2\mathbf{a}_t(X) + O_{L^2}(h^4), \\ W_2((E_{t,h})_\# p_t, p_{t+h}) &\le \|E_{t,h}(\Psi_t^{h^2/2}X) - F_{t,t+h}(X)\|_{L^2} \\ &\le \frac{L_aC_a}{8}h^4 + \frac{LC_a}{2}h^3 + \frac{C_j}{6}h^3.\end{aligned} \tag{29}$$

The shift supplies $\tau\mathbf{a}_t(X)$, which is the second-order displacement in the exact expansion. It also changes the point at which Euler evaluates velocity, producing the difference $h[u_t(\Psi_t^\tau X) - u_t(X)]$. Since the shift is $O(h^2)$ in root mean square, the spatial Lipschitz bound makes this additional displacement $O(h^3)$.

This local estimate starts from the exact distribution. In a sequence of Euler steps, we must also bound error from earlier steps. For a Lipschitz map E and measures $\mu, \nu \in \mathcal{P}_2$, apply E to both members of a coupling and take the infimum over couplings. This gives $W_2(E_\#\mu, E_\#\nu) \le \mathrm{Lip}(E)W_2(\mu,\nu)$. For Euler, $\mathrm{Lip}(E_{t,h}) \le 1 + Lh$. With $e_k = W_2(\widehat{p}_k, p_{t_k})$ and $C_* = C_j/6 + LC_a/2 + TL_aC_a/8$, the triangle inequality separates the earlier error from the new local error. Since $h_k \le T$,

$$\begin{aligned}e_{k+1} &\le W_2((E_{t_k,h_k})_\#\widehat{p}_k, (E_{t_k,h_k})_\# p_{t_k}) \\ &\quad + W_2((E_{t_k,h_k})_\# p_{t_k}, p_{t_{k+1}}) \\ &\le (1+Lh_k)e_k + C_*h_k^3.\end{aligned}$$

Starting from $e_0 = 0$, we apply this inequality at each step. Later steps increase each local error by at most $\prod_j(1+Lh_j) \le e^{LT}$. The sum satisfies $\sum_j h_j^3 \le H^2\sum_j h_j = TH^2$, which gives the claimed order. □

If the initial numerical distribution differs from p_0, the same proof adds $e^{LT}W_2(\widehat{p}_0, p_0)$ to the right side.

The modified equation. The modified equation produces the same cancellation by changing the velocity field. For a fixed step parameter h, let Y_s solve $\dot{Y}_s = u(t+s, Y_s) - (h/2)\mathbf{a}(t+s, Y_s)$ from $Y_0 = x$ over $0 \le s \le h$. Physical time evolves while h remains fixed. On compact sets, $u \in C^2$ and $\mathbf{a} \in C^1$ give $Y_s = x + su(t,x) + O(hs + s^2)$, and substitution yields

$$u(t+s, Y_s) = u(t,x) + s\mathbf{a}(t,x) + O(hs + s^2), \qquad \mathbf{a}(t+s, Y_s) = \mathbf{a}(t,x) + O(s).$$

After integration, the terms $h^2\mathbf{a}(t,x)/2$ cancel, giving $Y_h = x + hu(t,x) + O(h^3)$. For sufficiently small h, the remainders are uniform on compact sets. This is the first field correction in backward error analysis (Hairer et al., 2006, Chapter IX). Under the theorem's global assumptions, $u - (h/2)\mathbf{a}$ has a unique global flow. The acceleration condition leaves the weak continuity equation unchanged, so uniqueness gives the same distribution as u at every time. The expansion on compact sets alone does not bound mean squared error for an unbounded source. The coupling proof gives the integrable estimates needed for the finite-time bound.

C A necessary local condition and terminal cancellation

If one Euler step from the exact distribution has error $o(h^2)$, the field must satisfy the acceleration condition at that time. Terminal accuracy on a family of grids does not imply this local condition, because errors from different times can cancel. We prove the local result and then give an example of terminal cancellation.

> **Proposition 4, restated.** Let u and p_t satisfy the smoothness, growth, Lipschitz, and moment assumptions of Theorem 3, without requiring zero weighted divergence of acceleration under the current distribution. At a fixed $t < T$, suppose
>
> $$W_2((E_{t,h})_{\#}p_t, p_{t+h}) = o(h^2) \quad \text{as } h \downarrow 0.$$
>
> Then $\mathbf{a}_t$ has zero weighted divergence under p_t.

Proof of Proposition 4. For $X \sim p_t$, compare one Euler step $Y = E_{t,h}X$ with exact motion $Y + \Delta X$, where $\Delta X = F_{t,t+h}X - Y = h^2\mathbf{a}_t(X)/2 + \rho_{t,h}$. Equation (28) gives $\|\rho_{t,h}\|_{L^2} = O(h^3)$ and $\|\Delta X\|_{L^2} = O(h^2)$. Since a smooth compactly supported test ϕ has bounded first and second derivatives, Taylor expansion gives

$$\mathbb{E}[\phi(Y) - \phi(Y + \Delta X)] = -\mathbb{E}[\nabla\phi(Y) \cdot \Delta X] + O(\mathbb{E}\|\Delta X\|^2).$$

Replacing ΔX by $h^2\mathbf{a}_t(X)/2$ changes the expectation by $O(h^3)$. Replacing $\nabla\phi(Y)$ by $\nabla\phi(X)$ changes it by at most $Ch^3\|u_t\|_{L^2(p_t)}\|\mathbf{a}_t\|_{L^2(p_t)}$. Hence

$$\mathbb{E}\phi(E_{t,h}X) - \mathbb{E}\phi(F_{t,t+h}X) = -\frac{h^2}{2}\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t + o(h^2). \tag{30}$$

The expectation difference has absolute value at most $\mathrm{Lip}(\phi)W_2((E_{t,h})_{\#}p_t, p_{t+h})$, which tends to zero after division by h^2. The acceleration integral is therefore zero for every smooth compactly supported test, which proves zero weighted divergence. The moment requirements are the finite second moment and the stated L^2 bounds on acceleration and jerk, with C_j controlling the trajectory remainder through Eq. (28). □

Conversely, under all assumptions and common bounds in Appendix B, zero weighted divergence of acceleration gives local $O(h^3)$ error. Both directions concern one step started from the exact distribution.

For a terminal counterexample, let $T = 1$, $p_0 = \mathcal{N}(0, I_d)$, and

$$u(t,x) = \sin(2\pi t)e_1.$$

Here e_1 is the first coordinate vector. The particles translate in this direction during the first half of the interval and return during the second half, giving zero exact terminal displacement. Euler also gives zero displacement on every uniform grid with $N \ge 1$ steps, since

$$\frac{1}{N}\sum_{k=0}^{N-1} \sin(2\pi k/N)e_1 = 0.$$

For $N > 1$, summing the Nth roots of unity and taking imaginary parts gives this identity, while $N = 1$ follows directly. However, $\mathbf{a}_t = 2\pi \cos(2\pi t) e_1$ has nonzero weighted divergence whenever it is nonzero. The current distribution is a translated Gaussian, whose density has a nonzero derivative along any nonzero constant direction. Thus terminal exactness does not imply the local condition.

C.1 Weak expansion with an explicit remainder

Weak error compares expectations of test functions. We calculate its leading term and give an explicit cubic remainder for C_b^3 tests. These tests have continuous bounded values and derivatives through order three. This larger test class requires a stronger moment assumption than the local converse. The expansion alone does not bound W_2. The coupling in Appendix B gives that bound under the assumptions stated there.

Proposition 13 (Local weak error of Euler)**.** *Fix t and $h_0 > 0$. Suppose u is C^2 on a neighborhood of $[t, t + h_0] \times \mathbb{R}^d$, and one constant C satisfies*

$$\|u(s,x)\| + \|\partial_s u(s,x)\| + \|\partial_s^2 u(s,x)\| \le C(1 + \|x\|),$$
$$\|D_x u(s,x)\|_{\mathrm{op}} + \|D_x \partial_s u(s,x)\|_{\mathrm{op}} + \|D_x^2 u(s,x)\|_{\mathrm{op}} \le C.$$

Let $X \sim p_t$ have finite third moment, let p_{t+h} be its exact evolved law, and let $\phi \in C_b^3(\mathbb{R}^d)$. Then

$$\mathbb{E}\phi(X + h u_t(X)) - \mathbb{E}_{p_{t+h}}\phi = -\frac{h^2}{2}\mathbb{E}_{p_t}[\nabla\phi \cdot \mathbf{a}_t] + R_h, \qquad |R_h| \le C_{\mathrm{weak}} h^3 \mathbb{E}(1 + \|X\|^3).$$

The constant C_{weak} depends on C, h_0, and the supremum norms of $\nabla\phi$, $D^2\phi$, and $D^3\phi$, but not on h.

Proof. The numerical and exact test expansions agree at first order. Gronwall's inequality bounds the exact position by a constant times one plus the source radius, while acceleration has at most linear growth. Its derivative along the exact flow is

$$\mathbf{j} = \partial_t^2 u + 2(D_x \partial_t u)u + D_x^2 u[u,u] + (D_x u)\partial_t u + (D_x u)^2 u.$$

This derivative has at most quadratic growth. Taylor's formula along the straight Euler step is

$$\phi(x + h u_t(x)) = \phi(x) + h\nabla\phi \cdot u_t + \frac{h^2}{2} D^2\phi[u_t, u_t] + O(h^3(1 + \|x\|^3)).$$

Along the exact flow the second derivative of the test is $D^2\phi[u,u] + \nabla\phi \cdot \mathbf{a}$, and its third derivative is $D^3\phi[u,u,u] + 3D^2\phi[u,\mathbf{a}] + \nabla\phi \cdot \mathbf{j}$. The derivative and flow bounds make this third derivative at most $C_{\mathrm{weak}}(1+\|x\|^3)$ in absolute value. Subtracting the expansions cancels the terms that depend only on velocity and leaves the stated acceleration term. The finite third moment makes the remainder integrable. □

For smooth tests with compact support, integration by parts expresses the leading difference between the two distributions as

$$(\mathrm{Id} + h u_t)_{\#} p_t - p_{t+h} = \frac{h^2}{2} \nabla \cdot (p_t \mathbf{a}_t) + O_{\mathrm{weak}}(h^3).$$

The equality holds after integration against smooth compactly supported tests, up to the stated weak remainder. Integration by parts changes the sign of the acceleration integral and gives the positive divergence coefficient. When acceleration has zero weighted divergence, cutoff approximation extends this cancellation to the bounded tests in the proposition because acceleration is integrable.

For quadratic tests, the Euler expansion stops at order two, while the third derivative along the exact flow grows at most cubically. The third-moment assumption therefore makes the same formula valid. For affine fields $u(s,x) = B(s)x + d_{\mathrm{aff}}(s)$ with bounded coefficient derivatives through order two, this growth is only quadratic, so a finite second moment suffices for quadratic tests, including mean and covariance entries.

For the fixed Gaussian correction, write $y = x - m_t$ and $B = a(I + \Omega)$. The time derivative $-B\xi$ cancels the constant part $B\xi$ in $(D_x u)u$. Thus

$$\mathbf{a}_{v_1} = B^2 y = 2a^2 \Omega y.$$

The matrix identities imply $\operatorname{tr}\Omega = 0$ and skew-symmetry of $\Sigma_t^{-1}\Omega$. Hence the Gaussian density satisfies

$$\nabla\cdot(p_t\Omega y) = p_t[\operatorname{tr}\Omega - y^\top\Sigma_t^{-1}\Omega y] = 0.$$

Acceleration is nonzero away from the mean, but its weighted divergence is zero. The leading weak error therefore cancels. The Gaussian covariance recursion gives the convergence rate directly, while the coupling theorem bounds distribution error for more general distributions. Proposition 4 shows that the local condition is necessary under the stated second-moment, acceleration, and jerk bounds.

D A nonlinear radial path and its exact error coefficients

For a nonlinear radial path, covariance is not enough to determine the distribution. We therefore calculate distribution error from the full distribution of radii. Radial symmetry gives an optimal coupling that matches angles and pairs radii in increasing order. We show that the numerical radius maps keep this order and calculate their error over all steps. Bounds valid at every source radius then let us take expectations of the error limits. To calculate particle error, we also include the angle difference between particles from the same starting point.

For Theorem 5, assume $a, \eta, T > 0$. Let p_0 be rotationally invariant on $\mathbb{R}^2$, with $m_{2,0} = \int \|x\|^2 p_0(\,\mathrm{d}x) \in (0,\infty)$. A density is not required. Define

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \alpha(r) = a + \eta\frac{r^2}{1+r^2}, \qquad \mathcal{D} = r\frac{\mathrm{d}}{\mathrm{d}r},$$
$$q_\alpha = \mathcal{D}\alpha, \qquad P = \alpha^2 + \alpha q_\alpha, \qquad \beta = \sqrt{P},$$

using the positive square root, and put

$$u_0(x) = \alpha(\|x\|)x, \qquad u_1(x) = \alpha(\|x\|)x + \beta(\|x\|)Jx. \tag{31}$$

Let $R(t,r)$ solve $\partial_t R = f(R)$, with $f(r) = \alpha(r)r$ and $R(0,r) = r$. Write $R_0 = \|X_0\|$ for $X_0 \sim p_0$.

Proof of Theorem 5. Both fields generate the same probability path. The correction increases action, but it also makes acceleration preserve the current distribution. We first compare the numerical and exact radii to show that the correction improves distribution accuracy on every grid. We then calculate the radius and angle errors over all steps. Their positive leading coefficients give the stated convergence orders, with mean square bounds that justify taking the limits.

D.1 Regularity, path preservation, and action

Velocity bounds. For positive radius, $\mathcal{D}$ differentiates with respect to log radius. This gives the radius-weighted derivatives needed to differentiate the fields in space. With $a_+ = a + \eta$ and $s = r^2$, direct differentiation gives

$$q_\alpha = \frac{2\eta s}{(1+s)^2}, \quad \mathcal{D}q_\alpha = \frac{4\eta s(1-s)}{(1+s)^3}, \quad \mathcal{D}^2 q_\alpha = \frac{8\eta s(1-4s+s^2)}{(1+s)^4}.$$

Thus

$$a \le \alpha \le a_+, \quad 0 \le q_\alpha \le \eta/2, \quad |\mathcal{D}q_\alpha| \le \eta, \quad |\mathcal{D}^2 q_\alpha| \le 2\eta, \quad \mathcal{D}q_\alpha + 2q_\alpha = \frac{8\eta s}{(1+s)^3} \ge 0.$$

We have $P \ge a^2$ and $\mathcal{D}P = \alpha(2q_\alpha + \mathcal{D}q_\alpha) + q_\alpha^2 \ge 0$, with strict positivity for $r > 0$. Thus β is nondecreasing and strictly increases at positive radii. The coefficients are smooth functions of r^2, so the Cartesian fields are smooth at the origin as well.

We next bound β and its derivatives using P and its first two $\mathcal{D}$ derivatives. These estimates will give spatial Lipschitz bounds valid at every radius. Define

$$P_* = a_+^2 + a_+\eta/2, \qquad P_1 = 2a_+\eta + \eta^2/4, \qquad P_2 = 4a_+\eta + 2\eta^2,$$
$$b_* = \sqrt{P_*}, \qquad b_1 = P_1/(2a), \qquad b_2 = P_2/(2a) + P_1^2/(4a^3).$$

The identity $\mathcal{D}^2 P = 2q_\alpha^2 + 2\alpha \mathcal{D} q_\alpha + 3q_\alpha \mathcal{D} q_\alpha + \alpha \mathcal{D}^2 q_\alpha$ implies $|\mathcal{D}P| \le P_1$ and $|\mathcal{D}^2 P| \le P_2$. Hence

$$\beta \le b_*, \qquad |\mathcal{D}\beta| \le b_1, \qquad |\mathcal{D}^2\beta| \le b_2.$$

The spatial derivative of either $k(r)x$ or $k(r)Jx$ has operator norm at most $|k| + |\mathcal{D}k|$. Thus $a_+ + \eta/2$ and $a_+ + \eta/2 + b_* + b_1$ are global Lipschitz constants for u_0 and u_1, respectively, and both fields have linear growth and global flows.

The common path and its changing shape. Both fields have radial velocity $f(r)$. The correction adds the angle change $\Theta(t, r) = \int_0^t \beta(R(v, r))\,\mathrm{d}v$, which depends on the starting radius. At each positive source radius, rotation preserves the uniform distribution of angles. Both flows also keep the origin fixed, including any mass there. The two fields therefore generate the same p_t at every time. Their common radius flow satisfies

$$e^{at} r \le R(t, r) \le e^{a_+ t} r, \qquad \partial_r R(t, r) = \exp\left(\int_0^t f'(R(v, r))\,\mathrm{d}v\right) > 0. \tag{32}$$

The distributions have finite second moments.

To show that the distribution changes shape, the source support must contain at least two distinct positive radii. For $r_2 > r_1 > 0$, scalar-flow ordering and strict increase of α imply

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \frac{R(t, r_2)/r_2}{R(t, r_1)/r_1} = \alpha(R(t, r_2)) - \alpha(R(t, r_1)) > 0.$$

Distinct positive source radii therefore expand by different ratios for $t > 0$. Increasing transport between radius distributions is unique up to zero source probability, so no single scale factor can give the same distribution. The path thus changes shape when the source support contains at least two distinct positive radii. A source concentrated on one positive radius does not give this conclusion.

Acceleration preservation and moment bounds. The angular correction preserves the path. Choosing $\beta^2 = P$ also cancels the radial component of vector acceleration, even though the scalar radius can have a nonzero second derivative. For example, $u(x) = (I + J)x$ gives $X_t = e^t(\cos t, \sin t)$ from $(1, 0)$, with $R'' = R$ but vector acceleration $2JX_t$. In polar coordinates its radial component is $R'' - R(\dot{\theta})^2 = 0$. For the nonlinear field,

$$\mathbf{a}_{u_1}(x) = (\alpha^2 + \alpha q_\alpha - \beta^2)x + (2\alpha\beta + \alpha\mathcal{D}\beta)Jx = \beta_a(r)Jx, \quad \beta_a = \alpha(2\beta + \mathcal{D}\beta) \ge 2a^2. \tag{33}$$

With physical time fixed, the acceleration flow keeps radius r fixed and changes the angle by $v\beta_a(r)$. This rotation preserves every rotationally invariant measure, including measures without a density. The expectation of each smooth test stays constant, so differentiation proves zero weighted divergence.

To verify the general theorem's remaining assumptions, put

$$C_{\beta_a,0} = a_+(2b_* + b_1), \qquad C_{\beta_a,1} = \eta b_* + (2a_+ + \eta/2)b_1 + a_+ b_2.$$

Then $|\beta_a| \le C_{\beta_a,0}$, $|\mathcal{D}\beta_a| \le C_{\beta_a,1}$, and $L_a = C_{\beta_a,0} + C_{\beta_a,1}$ is a global Lipschitz bound for acceleration. The derivative of acceleration along the exact flow is

$$\mathbf{j}_{u_1}(x) = -\beta_a\beta x + \alpha(\beta_a + \mathcal{D}\beta_a)Jx.$$

The moment bounds required by the theorem are

$$C_a \le C_{\beta_a,0} e^{a_+ T}\sqrt{m_{2,0}}, \qquad C_j \le [b_* C_{\beta_a,0} + a_+(C_{\beta_a,0} + C_{\beta_a,1})]e^{a_+ T}\sqrt{m_{2,0}}.$$

These require only the source second moment.

Action comparison. The base field has potential

$$\Phi(x) = \frac{a+\eta}{2}\|x\|^2 - \frac{\eta}{2}\log(1+\|x\|^2), \qquad u_0 = \nabla\Phi.$$

The potential has at most quadratic growth and its gradient has at most linear growth, so the cutoff argument in Appendix H applies. By Eq. (32), all later radii are at most a fixed multiple of the source radius. The finite source second moment then gives one integrable bound on the squared cutoff error. We can therefore take the limit after expectation and time integration. For every finite-action field v satisfying the same weak continuity equation,

$$\mathcal{A}_T(v) = \mathcal{A}_T(u_0) + \int_0^T \|v_t - u_0\|^2_{L^2(p_t)}\,\mathrm{d}t.$$

The integral is nonnegative, so the base field minimizes action. Equality holds only when the fields agree almost everywhere. If the density is positive everywhere and the competing fields are continuous, they agree at every point. For the corrected field, the radial and rotational velocities are perpendicular at every point. Thus

$$\mathcal{A}_T(u_1) - \mathcal{A}_T(u_0) = \int_0^T \mathbb{E}[P(R(t,R_0))R(t,R_0)^2]\,\mathrm{d}t > 0. \tag{34}$$

Both actions are finite. In particular, $\mathcal{A}_T(u_1) \le T(a_+^2 + P_*)e^{2a_+T}m_{2,0}$.

D.2 Exact radial transport and improvement on every grid

The general theorem gives a second-order upper bound. To compare the errors on every grid, we use the order of the exact and numerical radii. One Euler step has radius maps

$$G_{0,h}(r) = r(1+h\alpha(r)), \qquad G_{1,h}(r) = r\sqrt{(1+h\alpha(r))^2 + h^2P(r)}.$$

Both maps preserve radius order. The base derivative is $1 + h(\alpha + q_\alpha) > 0$, while the corrected square-root factor is positive and nondecreasing, so its product with r has positive derivative.

Define

$$\mathcal{K} = \alpha^3 + 3\alpha^2 q_\alpha + \alpha q_\alpha^2 + \alpha^2 \mathcal{D}q_\alpha.$$

The exact derivatives are $R'' = PR$ and $R''' = \mathcal{K}R$. Since $\mathcal{D}q_\alpha \ge -2q_\alpha$, $\mathcal{K} \ge \alpha^3 + \alpha^2 q_\alpha + \alpha q_\alpha^2 \ge a^3 > 0$. Taylor's integral formula implies $R(h,r) > r + hf(r) + h^2P(r)r/2$ for $r, h > 0$. Also,

$$\left(1 + h\alpha + \frac{h^2P}{2}\right)^2 - [(1+h\alpha)^2 + h^2P] = h^3\alpha P + \frac{h^4P^2}{4} > 0.$$

The corrected map lies below that quadratic polynomial. Therefore

$$G_{0,h}(r) < G_{1,h}(r) < R(h,r) \quad (r,h>0). \tag{35}$$

Let $R_{i,k}$ compose the numerical radius maps through node k. Induction using monotonicity and the exact flow property proves

$$R_{0,k}(r) < R_{1,k}(r) < R(t_k, r) \quad (r > 0,\ k \ge 1).$$

All three maps send zero to zero.

The norm map is 1-Lipschitz, so spatial W_2 is at least the one-dimensional distance between the radius distributions. To achieve this lower bound, pair radii at the same quantile and give both particles the same independent uniform angle. All radius maps increase with the source radius. Using the same source radius therefore gives this quantile pairing, even when a radius has positive probability. Numerical angle changes depend only on radius and preserve radial symmetry. Thus

$$W_2(\widehat{p}_k^i, p_{t_k})^2 = \mathbb{E}[(R_{i,k}(R_0) - R(t_k, R_0))^2], \qquad i = 0, 1. \tag{36}$$

Positive radii have positive probability since $m_{2,0} > 0$. The strict pointwise inequalities prove

$$0 < W_2(\widehat{p}_k^1, p_{t_k}) < W_2(\widehat{p}_k^0, p_{t_k}) \quad (k \ge 1)$$

on every finite increasing grid, without a small-step assumption.

D.3 The three positive leading coefficients

With a, η, T, p_0 fixed, we refine uniform steps $h = T/N$ and calculate the radius error over all steps. We use the following quantities for the local radius errors and their effect at the final time:

$$\begin{aligned} V(T,s;r) &= \exp\left(\int_s^T f'(R(v,r))\,\mathrm{d}v\right), \\ d_0(r) &= \tfrac{1}{2}P(r)r, \\ d_1(r) &= r\left(\tfrac{1}{2}\alpha(r)P(r) + \tfrac{1}{6}\mathcal{K}(r)\right), \\ \mathcal{C}_i(T,r) &= \int_0^T V(T,s;r)d_i(R(s,r))\,\mathrm{d}s \quad (i=0,1), \\ \Theta_1(T,r) &= \tfrac{1}{2}\int_0^T \beta_a(R(s,r))\,\mathrm{d}s. \end{aligned}$$

The coefficient d_i gives the local radius error, and V describes how much the remaining exact flow increases it. Integrating their product gives $\mathcal{C}_i$, the leading radius-error coefficient at the final time. The angle-error coefficient is Θ_1. The limits below are positive, so the errors have the orders stated in Theorem 5:

$$\lim_{N\to\infty} \frac{W_2(\widehat{p}_N^0, p_T)}{h} = \|\mathcal{C}_0(T,R_0)\|_{L^2} > 0, \tag{37}$$

$$\lim_{N\to\infty} \frac{W_2(\widehat{p}_N^1, p_T)}{h^2} = \|\mathcal{C}_1(T,R_0)\|_{L^2} > 0, \tag{38}$$

$$\lim_{N\to\infty} \frac{\|\widehat{X}_N^1 - X_T^1\|_{L^2}}{h} = \|R(T,R_0)\Theta_1(T,R_0)\|_{L^2} > 0. \tag{39}$$

The last line pairs particles with the same start under the same corrected field.

Radius limits. For a fixed source radius, Taylor expansion gives

$$R(h,r) - G_{0,h}(r) = h^2 d_0(r) + O(h^3), \qquad R(h,r) - G_{1,h}(r) = h^3 d_1(r) + O(h^4).$$

The corrected cubic term is $-\alpha P r h^3/2$. The exact cubic term is $\mathcal{K} r h^3/6$. Their difference is d_1 times the cubed step length.

For fixed $r > 0$, the exact and numerical radii remain in a compact interval over $[0,T]$. With $\Delta_{i,k} = R(t_k,r) - R_{i,k}(r)$, the local bounds and discrete Gronwall inequality give $\Delta_{0,k} = O(h)$ and $\Delta_{1,k} = O(h^2)$ uniformly in k. Scaling by $z_k = \Delta_{i,k}/h^{i+1}$ and expanding the one-step flow in its starting radius gives

$$z_{k+1} = z_k + h[f'(R(t_k,r))z_k + d_i(R(t_k,r))] + O(h^2).$$

For this fixed radius, the remainder is bounded uniformly in k. The recurrence is Euler's method for $z' = f'(R(t,r))z + d_i(R(t,r))$, $z(0) = 0$, with a remainder of order h^2 at each step. There are T/h steps, and the factors that increase earlier errors are bounded. Discrete Gronwall therefore shows that the total remainder tends to zero. The scaled radius error converges to the solution of this linear equation. Variation of constants gives its terminal value $\mathcal{C}_i(T,r)$.

To pass the limit through expectation, we need a bound proportional to r. Its square is integrable because the source has a finite second moment. For this rational family, every fixed-order derivative $\mathcal{D}^m\alpha$ is bounded, and $P \geq a^2$ gives the same property for $\mathcal{D}^m\beta$. With $H_1 = \alpha$ and $H_{m+1} = \alpha(H_m + \mathcal{D}H_m)$, the exact time derivatives satisfy

$$\partial_t^m R(t,r) = R(t,r)H_m(R(t,r)), \qquad |\partial_t^m R(t,r)| \leq e^{a_+ h_0}\|H_m\|_\infty r \quad (0 \leq t \leq h_0,\ m \leq 4).$$

The remainder in the exact cubic expansion is therefore bounded by $Ch^4 r$ uniformly in r. For the numerical map, set $Q_* = a_+^2 + P_*$ and $M_* = 2a_+ + 2Q_* h_0$. Since $D_h = 1 + 2h\alpha + h^2(\alpha^2 + P) \geq 1$ for $h \geq 0$,

$$\left|\partial_h^4 \sqrt{D_h}\right| \leq \frac{15}{16}M_*^4 + \frac{9}{2}Q_* M_*^2 + 3Q_*^2.$$

The numerical cubic remainder is therefore also bounded by $Ch^4 r$, giving local radius-error bounds $Ch^2 r$ and $Ch^3 r$. The bounds on $\mathcal{D}\alpha$ and $\mathcal{D}P$ give $|\partial_r G_{i,h}| \le 1 + Ch$ for $0 \le h \le h_0$. A local error at time t_k is at most $Ch^{i+2} e^{a_+ t_k} r$, and later steps increase it by at most a factor of e^{CT}. Summing over $N = T/h$ steps gives

$$0 \le \Delta_{0,k}(r) \le C_T h r, \qquad 0 \le \Delta_{1,k}(r) \le C_T h^2 r. \tag{40}$$

Both scaled squared errors are bounded by $C_T^2 r^2$, which is integrable under the source second moment. Dominated convergence then lets us pass the limit through the expectation in the transport formula. This proves both distribution limits.

Particle limit. The optimal distribution coupling matches angles. Particles from the same starting point have an angle difference as well. One numerical step changes the angle by

$$\arctan \frac{h\beta(r)}{1 + h\alpha(r)} = h\beta(r) - h^2 \alpha(r)\beta(r) + O(h^3).$$

The exact increment from the same radius is

$$\int_0^h \beta(R(s,r))\,\mathrm{d}s = h\beta(r) + \frac{h^2}{2}\alpha(r)\mathcal{D}\beta(r) + O(h^3).$$

The exact increment minus the numerical increment equals $h^2 \beta_a(r)/2 + O(h^3)$. The angle remainders are uniform in r. For the exact angle, its third derivative has absolute value at most $a_+(\eta/2)b_1 + a_+^2 b_2$. For the numerical angle, write the increment as $\operatorname{Im} \log(1 + h[\alpha + \mathrm{i}\beta])$. Its third h derivative has absolute value at most $2(a_+^2 + b_*^2)^{3/2}$, because $|1 + h(\alpha + \mathrm{i}\beta)| \ge 1$. Taylor's formula therefore bounds the difference remainder by

$$\left[\frac{a_+(\eta/2)b_1 + a_+^2 b_2}{6} + \frac{(a_+^2 + b_*^2)^{3/2}}{3} \right] h^3$$

for every radius.

At each step, the method evaluates angular velocity at the numerical radius. By Eq. (40), this radius has error $O(h^2 r)$. Both exact and numerical radii remain at least as large as their positive source radius. The bound $|\beta(r_1) - \beta(r_2)| \le b_1 |\log(r_1/r_2)|$ therefore shows that using the exact radius instead changes one angle increment by $O(h^3)$. This estimate is uniform over positive source radii. We sum the local differences and use the bounded derivative of $\beta_a(R(t,r))$ to replace the sum by an integral. This gives

$$\Theta(T,r) - \Theta_N(r) = h\Theta_1(T,r) + O(h^2)$$

uniformly for $r > 0$.

The squared distance between the particles at the final time is

$$(R_{1,N} - R_T)^2 + 4R_{1,N}R_T \sin^2 \frac{\Theta_N - \Theta_T}{2}, \qquad R_T = R(T,r).$$

After division by h^2, the radial term tends to zero and the angular term tends to $R_T^2 \Theta_1(T,r)^2$. The radius and angle bounds show that the scaled squared distance is at most Cr^2. The distance is zero at $r = 0$, so dominated convergence proves the particle limit.

Since $f' = \alpha + q_\alpha \ge a$, $d_0(r) \ge a^2 r/2$, $d_1(r) \ge 2a^3 r/3$, and $\beta_a \ge 2a^2$,

$$\mathcal{C}_0(T,r) \ge \frac{a^2 T}{2} e^{aT} r, \quad \mathcal{C}_1(T,r) \ge \frac{2a^3 T}{3} e^{aT} r, \quad \Theta_1(T,r) \ge a^2 T.$$

The upper bounds make all three norms finite. Since $m_{2,0} > 0$, the norms are strictly positive, so the errors have the orders stated in Theorem 5. Setting $\eta = 0$ gives the isotropic Gaussian coefficients. The transport statements still hold for every radial source with finite second moment. □

D.4 Sensitivity to a fixed coefficient error

For fixed $\lambda \geq 0$, let $u_\lambda(x) = \alpha(r)x + \lambda\beta(r)Jx$. Each such field generates the same continuous path, and

$$\mathbf{a}_{u_\lambda} = (1-\lambda^2)P(r)x + \lambda\beta_a(r)Jx.$$

The decomposition required for the residual bound is $w_\lambda = \lambda\beta_a(r)Jx$ and $g_\lambda = (1-\lambda^2)P(r)x$. The tangential part is globally Lipschitz and preserves the current distribution, so it has zero weighted divergence. The radial residual obeys

$$\sup_t \|g_\lambda\|_{L^2(p_t)} \leq |1-\lambda^2| P_* e^{a_+ T}\sqrt{m_{2,0}}.$$

The same bounded radial derivatives give the remaining regularity and moment bounds, uniformly for λ in a fixed bounded interval. The residual estimate therefore gives $O(H^2 + |1-\lambda^2|H)$.

For $\lambda \neq 1$, the error itself has a positive first-order coefficient. To see this, expand the radius error:

$$R(h,r) - r\sqrt{(1+h\alpha(r))^2 + h^2\lambda^2 P(r)} = (1-\lambda^2)d_0(r)h^2 + O(h^3).$$

Since α and P are nondecreasing, the numerical radius map is increasing, so the same optimal transport pairing applies. The error recurrence has local coefficient $(1-\lambda^2)d_0$. The signed scaled error therefore tends to $(1-\lambda^2)\mathcal{C}_0(T,r)$. Its absolute value is bounded by $C_T r$, so the radial transport formula and dominated convergence give

$$\lim_{N\to\infty} \frac{W_2(\widehat{p}_N^\lambda, p_T)}{h} = |1-\lambda^2| \|\mathcal{C}_0(T,R_0)\|_{L^2} > 0. \tag{41}$$

Every fixed mismatch in the nonnegative coefficient therefore gives first-order distribution error. This limiting rate can be difficult to see over a finite range of small errors. If negative coefficients are allowed, both $\lambda = \pm 1$ cancel the leading term.

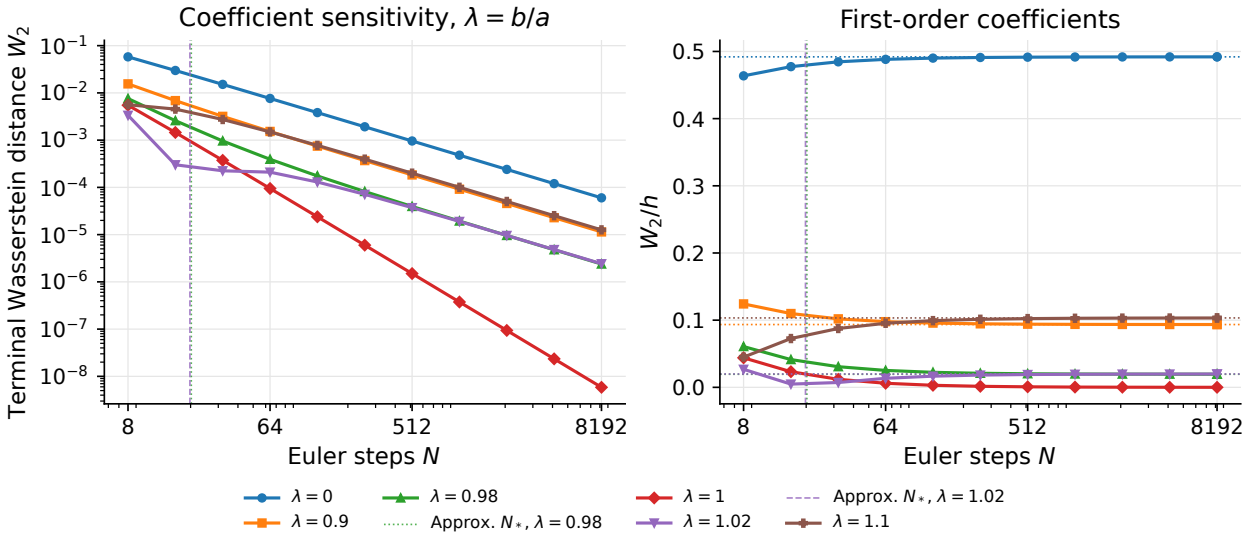


Figure 7: A fixed rotation mismatch gives first-order Euler distribution error for these Gaussian cases, with the parameters of Figure 2. Each curve keeps $\lambda = b/a$ fixed as the grid is refined. The right panel shows error divided by h. For $\lambda \neq 1$, this ratio tends to the positive value $|1-\lambda^2|c_0^{\mathrm{G}}$ shown by a horizontal dotted line. Section 7 defines c_0^{G}. With matched rotation $\lambda = 1$, the limit is zero. Vertical lines near $N = 20$ estimate where the first- and second-order terms have similar size for $\lambda = 0.98$ (dotted) and 1.02 (dashed). Markers distinguish the error curves from their limits.

D.5 Affine changes of coordinates, product distributions, and a dimension restriction

A fixed affine change of coordinates can give distributions that are not radial in the original coordinates. For $\mathcal{T}(x) = d_{\text{aff}} + Gx$ with G invertible, set $\widetilde{u}(t, y) = Gu(t, G^{-1}(y - d_{\text{aff}}))$ and $\widetilde{p}_t = \mathcal{T}_{\#} p_t$. Substitution into the differential equation and Euler update gives

$$\widetilde{F}_{s,t} \circ \mathcal{T} = \mathcal{T} \circ F_{s,t}, \qquad \widetilde{E}_{t,h} \circ \mathcal{T} = \mathcal{T} \circ E_{t,h}.$$

The transformed acceleration is $\widetilde{\mathbf{a}}_t(\mathcal{T}(x)) = G\mathbf{a}_t(x)$. For every compact smooth test ϕ,

$$\int \nabla\phi(y) \cdot \widetilde{\mathbf{a}}_t(y)\, \widetilde{p}_t(\,\mathrm{d}y) = \int \nabla(\phi \circ \mathcal{T})(x) \cdot \mathbf{a}_t(x)\, p_t(\,\mathrm{d}x) = 0.$$

Because the affine map is invertible, $\phi \circ \mathcal{T}$ has compact support. The map also preserves the required smoothness, Lipschitz, growth, and moment assumptions, with different constants.

Write $s_{\min}(G) > 0$ for the smallest singular value and $\|G\|_{\text{op}}$ for the operator norm. Push forward any coupling to obtain the upper bound below, then apply that bound to $\mathcal{T}^{-1}$ to obtain the lower bound:

$$s_{\min}(G) W_2(\mu, \nu) \le W_2(\mathcal{T}_{\#}\mu, \mathcal{T}_{\#}\nu) \le \|G\|_{\text{op}} W_2(\mu, \nu).$$

Differences between particles from the same starting point satisfy the same norm bounds. These factors are fixed, so the positive limits in Theorem 5 give orders $\Theta(h)$ and $\Theta(h^2)$ after the change of coordinates as well. The corrected distribution error is therefore smaller on sufficiently fine uniform grids. The estimates do not give exact coefficients or improvement on every grid. For example, $G = \operatorname{diag}(2, 1)$ gives an elliptical mixture whose shape changes with time. This example is still an affine image of the symmetric one. Since $G(Du_0)G^{-1}$ need not be symmetric, the transformed base field need not minimize kinetic action.

We can also add independent stationary coordinates. This gives the product path $p_t \otimes \nu_{\text{stat}}$ with field $(u_i, 0)$, where ν_{stat} has finite second moment. Projection onto the moving coordinates shows that the product W_2 is at least the original distance. Pairing each stationary coordinate with itself achieves this lower bound. Thus

$$W_2(\widehat{p}^i_k \otimes \nu_{\text{stat}}, p_{t_k} \otimes \nu_{\text{stat}}) = W_2(\widehat{p}^i_k, p_{t_k}).$$

The stationary coordinates also leave particle error unchanged. Taking ν_{stat} on $\mathbb{R}$ gives a three-dimensional example, and adding more coordinates gives examples in every dimension $d \ge 2$. Independent products of moving planar examples also retain both error orders and strict improvement, since their squared W_2 distances add across components.

A dimension restriction applies if a tangent correction must cancel radial acceleration everywhere on a sphere while keeping the given radial coefficient. For $u(x) = \alpha(r)x + V(x)$ with $x \cdot V(x) = 0$, differentiation in direction V gives $x \cdot (DV)V = -\|V\|^2$. Differentiating the remaining acceleration terms yields

$$x \cdot \mathbf{a}_u(x) = P(r)r^2 - \|V(x)\|^2.$$

Cancellation requires $\|V(x)\|^2 = P(r)r^2 > 0$ at every point of each positive-radius sphere. In odd-dimensional space, such a sphere has even surface dimension, and the hairy-ball theorem requires every continuous tangent field to vanish somewhere (Hatcher, 2002, Theorem 2.28). This rules out a tangent correction with the required cancellation at every point. Other fields can still have zero weighted divergence of acceleration, as the example with stationary coordinates shows.

E Robustness to an acceleration residual

When only part of acceleration preserves the distribution, we shift the starting point along that part. The remaining acceleration contributes a first-order term to the global error bound. Proposition 10 requires a bound on this remainder at every time, including Euler's evaluation times.

Proposition 10, restated. Let $T > 0$, $p_0 \in \mathcal{P}_2$, and $p_t = (F_{0,t})_{\#}p_0$. Suppose u is C^2 on a neighborhood of $[0,T]\times\mathbb{R}^d$, globally Lipschitz in space with constant L, and bounded at zero uniformly in time. Define $\mathbf{a}$ and $\mathbf{j}$ as in Appendix B, and assume $C_j = \sup_t \|\mathbf{j}_t\|_{L^2(p_t)} < \infty$. Suppose

$$\mathbf{a}_t = w_t + g_t, \qquad C_{w,2} = \sup_t \|w_t\|_{L^2(p_t)} < \infty, \qquad \sup_t \|g_t\|_{L^2(p_t)} \le \varepsilon.$$

For each time, require $w_t \in C^1$ to be globally Lipschitz with a common constant L_w and to have zero weighted divergence under p_t. With Euler's method and the grid defined above, put $C_w = C_j/6 + LC_{w,2}/2 + TL_wC_{w,2}/8$. Then

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT}\left(C_w \sum_j h_j^3 + \frac{\varepsilon}{2}\sum_j h_j^2\right) \le e^{LT}T\left(C_w H^2 + \frac{\varepsilon}{2}H\right). \tag{42}$$

Proof of Proposition 10. For $X \sim p_t$, hold physical time fixed and follow the flow of w_t for auxiliary duration $h^2/2$. This motion preserves p_t. The shift has L^2 norm at most $C_{w,2}h^2/2$ and remainder at most $L_wC_{w,2}h^4/8$ after its linear term. It supplies the w_t part of the exact second-order displacement $h^2(w_t+g_t)/2$. The residual and exact-flow remainder stay in the error. The paired outputs therefore have root mean squared distance at most

$$\frac{C_j}{6}h^3 + \frac{LC_{w,2}}{2}h^3 + \frac{L_wC_{w,2}}{8}h^4 + \frac{\varepsilon}{2}h^2.$$

The Lipschitz estimate from the main proof bounds how much later Euler steps increase each error. Summing these contributions and using $\sum_j h_j^3 \le TH^2$ and $\sum_j h_j^2 \le TH$ proves the result. □

A time-dependent residual contributes $\frac{1}{2}\sum_j h_j^2\varepsilon_{t_j}$. A bound on its time integral cannot control this sum, because the residual may be large at Euler's evaluation times. A small density-weighted divergence norm also need not give the required L^2 decomposition. Weighted L^2 projection separates acceleration into a part with zero weighted divergence and a remainder. This controls the norm, but it does not give the global Lipschitz bound required for the part that preserves the distribution.

The proposition bounds discretization error relative to the exact probability path of the approximate field. That path may differ from the intended path, so we must also estimate the error between the two paths. At the terminal time,

$$W_2(\widehat{p}_T^{\,\text{approx}}, p_T) \le W_2(\widehat{p}_T^{\,\text{approx}}, p_T^{\text{approx}}) + W_2(p_T^{\text{approx}}, p_T).$$

F Dependence on the numerical solver

Euler and Heun require different rotations to cancel their leading Gaussian distribution errors. We first compare their expansions at a common step length, then pair two corrected Euler steps with one base-field Heun step to compare equal numbers of field evaluations.

Proposition 14 (Euler and Heun for a fixed Gaussian rotation)**.** *Let $T > 0$, $a > 0$, and let $\Sigma_0 = C_0C_0^\top$ be positive definite in even dimension. Let $J^\top = -J$, $J^\top J = I$, $\Omega = C_0JC_0^{-1}$, $m_t = m_0 + t\xi$, and $p_t = \mathcal{N}(m_t, e^{2at}\Sigma_0)$. For a fixed real b, use*

$$v_b(t,x) = \xi + (aI + b\Omega)(x - m_t).$$

On uniform grids with $h = T/N$, Euler satisfies

$$W_2(\widehat{p}_N^{\,\mathrm{E},b}, p_T) = \sqrt{\operatorname{tr}\Sigma_0}\,e^{aT}\frac{T|a^2 - b^2|}{2}h + O(h^2). \tag{43}$$

Define Heun by

$$x^* = x + hv_b(t,x), \qquad x^+ = x + \frac{h}{2}[v_b(t,x) + v_b(t+h, x^*)].$$

Its terminal error satisfies

$$W_2(\widehat{p}_N^{\mathrm{H},b}, p_T) = \sqrt{\operatorname{tr}\Sigma_0}\, e^{aT}\, \frac{T|a(a^2-3b^2)|}{6} h^2 + O(h^3). \tag{44}$$

In particular, Heun's positive leading constant at $b = a$ is twice its constant at $b = 0$. The leading Heun term cancels at $b = \pm a/\sqrt{3}$, giving an $O(h^3)$ bound in this family.

Proof idea. In centered whitened coordinates, each solver acts as a polynomial in $a + \mathrm{i}b$. The real part of the polynomial's logarithm gives the numerical log scale. We use this scale to calculate Gaussian distribution error.

Proof. The exact centered flow $e^{at}C_0 e^{btJ}C_0^{-1}$ has the same covariance and mean for every b. Both solvers reproduce the affine mean exactly. For the centered whitened operator $B = aI + bJ$, Euler applies $I + hB$, while Heun applies $I + hB + h^2B^2/2$.

A matrix $cI + dJ$ multiplies whitened covariance by $c^2 + d^2$, the squared complex modulus of $c + \mathrm{i}d$. Thus scalar complex calculations give the distribution error. With $w = h(a + \mathrm{i}b)$, the exact, Euler, and Heun scale factors are e^{ah}, $|1 + w|$, and $|1 + w + w^2/2|$. Writing $\Delta^{\log}$ for exact minus numerical log scale after N steps, we use

$$\log(1 + w) = w - \tfrac{1}{2}w^2 + O(w^3),$$
$$\log(1 + w + w^2/2) = w - \tfrac{1}{6}w^3 + O(w^4).$$

Taking real parts and multiplying by $N = T/h$ shows that

$$\Delta_{\mathrm{E}}^{\log} = \frac{T}{2}(a^2 - b^2)h + O(h^2), \qquad \Delta_{\mathrm{H}}^{\log} = \frac{T}{6}a(a^2 - 3b^2)h^2 + O(h^3).$$

By the Gaussian scale identity, $W_2 = \sqrt{\operatorname{tr}\Sigma_0}\, e^{aT}|1 - e^{-\Delta^{\log}}|$. Substitution gives the stated errors. For Heun, the absolute cubic coefficients are $a^3/6$ at $b = 0$ and $a^3/3$ at $b = a$, so Euler's correction doubles Heun's leading coefficient. The cubic coefficient instead vanishes at $b^2 = a^2/3$. □

Euler's leading term cancels at $b = \pm a$, or only at $b = a$ if $b \geq 0$ is required. For fixed δ_b with $b = a(1 + \delta_b)$ and $b^2 \neq a^2$, Eq. (43) gives the positive first-order coefficient $\sqrt{\operatorname{tr}\Sigma_0}\, e^{aT} T a^2 |2\delta_b + \delta_b^2|/2$. Appendix D.4 gives the corresponding result for nonlinear fields.

For this family, applying the matrix exponential of the full constant centered operator gives the exact solution. An exponential integrator that splits or approximates this operator needs a separate error analysis.

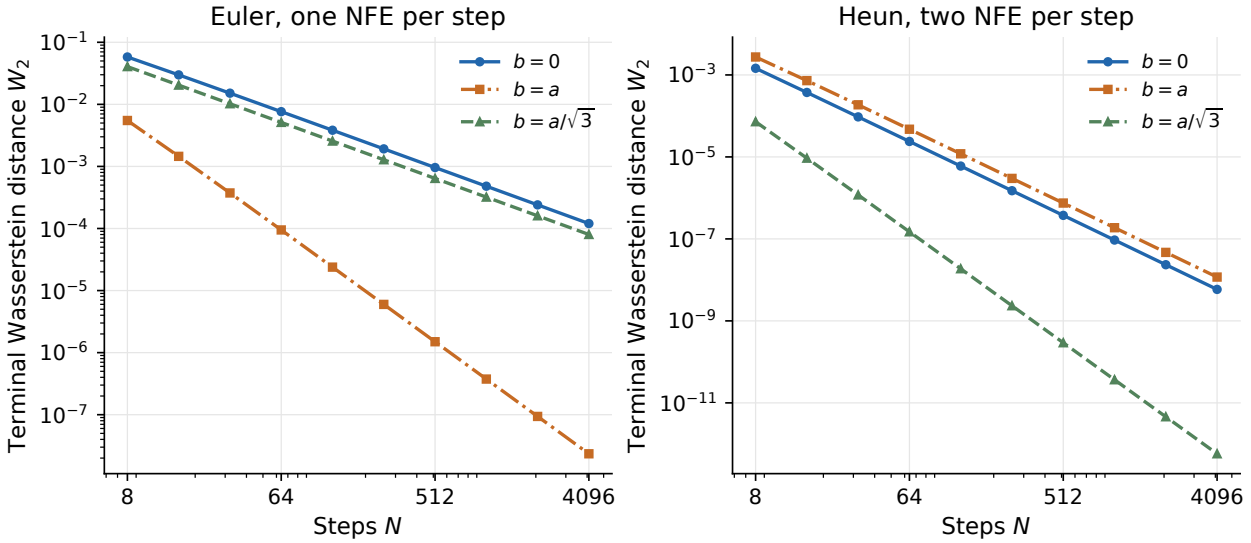


Figure 8: Euler and Heun require different rotations to cancel their leading distribution errors. Both methods use the Gaussian path and parameters of Figure 2, with the same number of uniform steps. Rotation $b = a$ improves Euler but doubles Heun's leading coefficient relative to $b = 0$. For Heun, $b = a/\sqrt{3}$ cancels the second-order term. Heun uses two evaluations per step, so the methods have unequal NFE here. Figure 6, Table 1, and Table 3 compare equal NFE.

F.1 Equal NFE for corrected Euler and base-field Heun

Proof of Proposition 9. Let N be even and $h = T/N$. In centered whitened coordinates, two corrected Euler steps of length h have scale multiplier

$$\left(\sqrt{1 + 2ah + 2a^2h^2}\right)^2 = 1 + a(2h) + \tfrac{1}{2}a^2(2h)^2.$$

This is also the positive scale multiplier of one base-field Heun step of length $2h$. Both methods reproduce the affine mean. Their covariances match, so their Gaussian distributions are identical at every shared node. Euler uses one evaluation per step and Heun uses two, giving N evaluations for each method on $[0, T]$. The same argument applies when consecutive Euler steps come in equal-length pairs on an otherwise nonuniform grid. □

F.2 Corrected Heun and base-field RK3 at equal NFE

At $b = a/\sqrt{3}$, Heun's second-order distribution term cancels and the remaining third-order coefficient is positive. We calculate this coefficient and compare Heun with base-field RK3 using the same number of evaluations. Put $\zeta = a + ib$, $R_2(w) = 1 + w + w^2/2$, and $C_\Sigma = \sqrt{\operatorname{tr}\Sigma_0}e^{aT}$. After $n = T/h$ uniform steps, Heun's errors are exactly

$$W_2^{\mathrm{H},b} = \sqrt{\operatorname{tr}\Sigma_0}\left|e^{aT} - |R_2(h\zeta)|^n\right|, \tag{45}$$

$$e_{\mathrm{p}}^{\mathrm{H},b} = \sqrt{\operatorname{tr}\Sigma_0}\left|R_2(h\zeta)^n - e^{T\zeta}\right|. \tag{46}$$

Here e_{p} is the root mean squared particle error from Section 2. The formulas also apply to elliptical covariance, since a map difference $cI + dJ$ in transformed coordinates contributes covariance $(c^2 + d^2)\Sigma_0$ in the original coordinates.

Since $\log R_2(w) = w - w^3/6 + w^4/8 + O(w^5)$, the choice $b = a/\sqrt{3}$ cancels the real cubic term. The remaining coefficients satisfy $\operatorname{Re}\zeta^4 = -8a^4/9$ and $\operatorname{Im}\zeta^3 = 8a^3/(3\sqrt{3})$. Thus

$$W_2^{\mathrm{H},a/\sqrt{3}} = C_\Sigma \frac{a^4 T}{9} h^3 + O(h^4), \tag{47}$$

$$e_{\mathrm{p}}^{\mathrm{H},a/\sqrt{3}} = C_\Sigma \frac{4a^3 T}{9\sqrt{3}} h^2 + O(h^3). \tag{48}$$

Both coefficients are positive, so distribution error is third order and particle error is second order.

For comparison, Kutta's three-stage, third-order Runge–Kutta method uses

$$k_1 = v(t,x), \quad k_2 = v(t + s/2, x + sk_1/2), \quad k_3 = v(t+s, x - sk_1 + 2sk_2), \quad x^+ = x + s(k_1 + 4k_2 + k_3)/6.$$

On the base field v_0, its centered scale factor is $R_3(as) = 1 + as + (as)^2/2 + (as)^3/6$. Because $\log R_3(z) = z - z^4/24 + O(z^5)$,

$$W_2^{\mathrm{RK3},0} = C_\Sigma \frac{a^4 T}{24} s^3 + O(s^4).$$

The base-field Runge–Kutta map is a positive scalar dilation. Pairing particles from the same starting point is therefore optimal, so particle error equals distribution error.

For an evaluation count N divisible by six, corrected Heun takes $N/2$ steps and base RK3 takes $N/3$. Substitution of $h = 2T/N$ and $s = 3T/N$ into the two error expansions proves

$$\lim_{N\to\infty} \frac{W_2^{\mathrm{H},a/\sqrt{3}}}{W_2^{\mathrm{RK3},0}} = \frac{64}{81}.$$

At the finite evaluation counts in Table 3, corrected Heun has 18.0–20.9% smaller distribution error, while its particle error is larger.

NFE	Corrected field Heun W_2	Base field, RK3 W_2 and particle error	Corrected field Heun particle error
12	1.677×10^{-4}	2.046×10^{-4}	3.904×10^{-3}
48	2.790×10^{-6}	3.497×10^{-6}	2.581×10^{-4}
192	4.426×10^{-8}	5.588×10^{-8}	1.636×10^{-5}
384	5.547×10^{-9}	7.011×10^{-9}	4.100×10^{-6}

Table 3: Corrected Heun has smaller distribution error but larger particle error than base-field RK3 at the listed evaluation counts. The parameters are those of Figure 2, with $b = a/\sqrt{3}$ for Heun and $b = 0$ for RK3. Both fields stay fixed as NFE increases.

G Restrictions on the mean, second moment, and action

The acceleration condition requires the mean to move at constant velocity. It also determines kinetic action from the path's second moment. If a path fails these restrictions, no field in the stated class can satisfy the acceleration condition. Other ways to improve numerical accuracy may still be possible.

For Proposition 6, let X_t follow a global flow on $[0,T]$. Assume its trajectories and their first two derivatives are continuous, with $\dot{X}_t = u_t(X_t)$ and $\ddot{X}_t = \mathbf{a}_t(X_t)$. One sufficient condition for the differentiations below is

$$\mathbb{E} \sup_{0\le t\le T} [\|X_t\|^2 + \|u_t(X_t)\|^2 + \|\mathbf{a}_t(X_t)\|^2] < \infty. \tag{49}$$

More generally, require that the second moment can be differentiated twice under the expectation and that $u_t, \mathbf{a}_t \in L^2(p_t)$. If $\mathbf{a}_t$ has zero weighted divergence under the current distribution, then

$$M''(t) = 2\mathbb{E}\|u_t(X_t)\|^2, \qquad M(t) = \mathbb{E}\|X_t\|^2. \tag{50}$$

Proof idea. The second derivative of the second moment contains kinetic energy and an acceleration term. To remove the acceleration term, we approximate squared distance by compactly supported tests and use zero weighted divergence. The remaining identity determines energy. This energy must be at least the infimum over fields satisfying the given continuity equation.

Proof of Proposition 6. Differentiating twice shows that

$$M''(t) = 2\mathbb{E}\|u_t(X_t)\|^2 + 2\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)].$$

Condition (49) gives an integrable derivative bound, so dominated convergence and Cauchy–Schwarz justify differentiation under expectation. To test zero weighted divergence, replace squared distance by $\chi_n(x)\|x\|^2/2$, using the smooth compact cutoffs of Appendix H. The gradient converges to x in $L^2(p_t)$ as the second-moment tail vanishes. Since $\mathbf{a}_t \in L^2(p_t)$, passage to the limit gives $\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)] = 0$ and proves Eq. (50).

Let $K_{\min}(t)$ be the infimum of kinetic energy over fields satisfying the weak continuity equation for the path at time t. These competitors need not satisfy the acceleration condition. The chosen field has energy at least this infimum, giving $M''(t) \geq 2K_{\min}(t)$. For a stationary distribution, $M'' = 0$ forces the field to vanish p_t-almost everywhere, or everywhere if the field is continuous and the density is positive throughout space.

For a centered radial dilation $p_t = (g(t)I)_{\#}p_0$, let p_0 be rotationally invariant and suppose $g > 0$ is C^2 and $M_0 = \int \|x\|^2 p_0(\mathrm{d}x) > 0$ is finite. The base field is the gradient $v_t(x) = (g'/g)x$. The cutoff argument in Appendix H places this gradient in the required closure. The projection identity then implies

$$K_{\min}(t) = M_0 g'(t)^2, \qquad M''(t) = 2M_0[g'(t)^2 + g(t)g''(t)].$$

When $g''(t) < 0$, the necessary energy inequality fails. No field with the stated regularity and finite energy can then have zero weighted divergence of acceleration. For a scaling path that does admit such a field, its kinetic energy above that of the base field must equal $M_0 g(t)g''(t)$. □

When $g'' = 0$, extra energy must be zero and the base acceleration is zero. If $g'' > 0$, an even-dimensional radial distribution can satisfy this necessary condition through

$$u(t,x) = \frac{g'}{g}x + b(t)Jx, \qquad b(t)^2 = \frac{g''}{g}, \quad J^\top = -J, \quad J^\top J = I.$$

Provided the square root has the required regularity, differentiation gives $\mathbf{a}_u = [b' + 2(g'/g)b]Jx$, which has zero weighted divergence for a radial distribution. Zeros of g'' require an additional check of square-root smoothness. Terminal accuracy can still occur when the local acceleration condition fails, through cancellation across steps (Appendix C).

Independent Gaussian interpolation at the midpoint. Independent interpolation pairs can have different kinetic energy from their conditional-mean field. For independent two-dimensional standard Gaussians X_0, X_1, let $X_t = (1-t)X_0 + tX_1$. At $t = 1/2$, the position depends on $X_0 + X_1$, which is independent of $X_1 - X_0$. The conditional mean velocity is therefore zero, although $\mathbb{E}\|X_1 - X_0\|^2 = 4$. The path remains nonstationary, with variance decreasing before the midpoint and increasing afterwards.

Proof of Corollary 7. The same integrability assumption also lets us differentiate the mean twice under expectation. By Condition (49),

$$m'(t) = \mathbb{E}u_t(X_t), \qquad m''(t) = \mathbb{E}\mathbf{a}_t(X_t).$$

To evaluate the acceleration expectation, choose a smooth compact cutoff χ equal to one near the origin and put $\chi_n(x) = \chi(x/n)$. Testing Eq. (4) with $\phi_n(x) = \chi_n(x)x_j$ gives gradients converging pointwise to e_j, with a bound independent of n. Since $\mathbf{a}_t \in L^2(p_t) \subset L^1(p_t)$, dominated convergence gives $\mathbb{E}\mathbf{a}_{t,j}(X_t) = 0$. Hence $m''(t) = 0$ and $m(t) = m(0) + t\mathbb{E}u_0(X_0)$.

Integrating Eq. (50) gives the action identity in Eq. (22). The moment assumption makes M twice continuously differentiable, with endpoint derivatives taken from inside the interval. Thus the given path determines both the energy at each time and the total action for every field in this class.

If a gradient base field v^∇ generates the given path and satisfies the conditions in Section 6, Eq. (23) gives

$$\int_0^T \|u_t - v_t^\nabla\|_{L^2(p_t)}^2 \, \mathrm{d}t = \tfrac{1}{2}\big[M'(T) - M'(0)\big] - \mathcal{A}_T(v^\nabla).$$

The path therefore determines the additional action within this class. The identity alone does not show that such a field exists.

The assumptions of Theorem 3 also imply the required moment condition. With $U = \sup_t \|u(t,0)\|$, Gronwall's inequality gives

$$\sup_{0\le t\le T} \|X_t\| \le (\|X_0\| + TU)e^{LT}.$$

Continuity gives $C_{a,0} = \sup_t \|\mathbf{a}_t(0)\| < \infty$, and the common Lipschitz bounds give $\|u_t(X_t)\| \le U + L\|X_t\|$ and $\|\mathbf{a}_t(X_t)\| \le C_{a,0} + L_a\|X_t\|$. Each squared time supremum is therefore bounded by $C(1 + \|X_0\|^2)$ for finite C, which is integrable under the source second moment. □

H Minimum action among fields with the same path

We compare actions by using a gradient base field that can be approximated in mean square by compactly supported gradients. Such a field is orthogonal to every correction that preserves the probability path. For compactly supported potentials, zero weighted divergence gives this orthogonality directly. For the unbounded Gaussian and radial potentials, we use cutoff approximation to obtain the same result in mean square.

We use $\mathcal{V}(p)$ for fields that are jointly continuous, have finite action, and are locally Lipschitz in space uniformly on compact time-space sets. Their flows must exist on $[0,T]$ and generate the given path. The action identity also holds for the larger class of finite-action fields satisfying the same weak continuity equation.

At each time, define the space of gradients and their mean square limits by

$$\mathcal{G}_t = \overline{\{\nabla\psi : \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(p_t)}.$$

Suppose $v_t \in \mathcal{G}_t$ for almost every time and v has finite action. Let r have finite action with $\nabla\cdot(p_t r_t) = 0$ distributionally for almost every time. For every smooth compactly supported potential, $\mathbb{E}_{p_t}[r_t \cdot \nabla\psi] = 0$. By continuity of the inner product in weighted $L^2(p_t)$, we also have $\mathbb{E}_{p_t}[r_t \cdot v_t] = 0$. The cross term is therefore zero when we expand squared velocity and integrate, proving Eq. (23). Equal action requires $r = 0$ almost everywhere under $dt\,p_t(dx)$, which means almost surely in position for almost every time. To obtain equality everywhere, we need the continuity and positivity argument below. Differences between fields satisfying the same continuity equation have the required zero weighted divergence. The existence of their particle flows is a separate requirement.

For the gradient fields used here, a finite second moment is enough to justify approximation by compactly supported gradients. Let $v_t(x) = \nabla\varphi_t(x)$, where φ_t has at most quadratic growth and its gradient has at most linear growth, uniformly on the fixed time interval. Choose a smooth cutoff function χ_n that is one on the radius-n ball, zero outside the radius-$2n$ ball, and has $\|\nabla\chi_n\| \le C/n$. Then

$$\nabla(\chi_n\varphi_t) - \nabla\varphi_t = (\chi_n - 1)\nabla\varphi_t + \varphi_t\nabla\chi_n.$$

The squared approximation error is bounded by a constant times $\mathbb{E}_{p_t}[(1+\|X\|^2)\mathbf{1}_{\{\|X\|>n\}}]$, which vanishes by the finite second moment. For radial dilation, $\|X_t\| = g_t\|X_0\|$ with bounded g_t gives one integrable tail bound for all times. The nonlinear path satisfies the same requirement through $\|X_t\| \le e^{(a+\eta)T}\|X_0\|$ in Eq. (32). Scaling and translating a fixed Gaussian gives the corresponding bound for the Gaussian path. Compactly supported gradients therefore converge in mean square both at each time and after time integration.

When every p_t has positive density throughout space, equal action in $\mathcal{V}(p)$ implies equality everywhere. A continuous difference nonzero at one point stays nonzero on an open time-space neighborhood. Every spatial ball in that neighborhood has positive p_t mass, contradicting almost-everywhere equality under the path. The Gaussian path therefore has a unique minimizer in this continuous class.

H.1 Gradient fields under the acceleration condition

Under all assumptions of Theorem 3, suppose each u_t is a gradient on $\mathbb{R}^d$. We prove that $\mathbf{a}_t = 0$ p_t-almost everywhere and that, for p_0-almost every start, Euler's method is exact on every finite increasing grid.

Proof of Proposition 8. A gradient velocity has a symmetric spatial Jacobian, so material acceleration is also a gradient. The growth bounds let us approximate acceleration by compactly supported gradients in the weighted norm. The acceleration condition makes acceleration orthogonal to all such gradients, so its norm must be zero.

Choose the normalized potential $\psi(t,x) = \int_0^1 u(t,sx)\cdot x\,\mathrm{d}s$. Because u_t is a gradient, $\nabla_x\psi = u_t$. The C^2 regularity of u lets us differentiate in time and space. Symmetry of $D_x u_t$ then implies

$$\mathbf{a}_t = \nabla q_{\mathrm{acc,t}}, \qquad q_{\mathrm{acc,t}} = \partial_t\psi_t + \tfrac{1}{2}\|\nabla\psi_t\|^2.$$

Subtract $q_{\mathrm{acc,t}}(0)$ from this potential. With $C_{a,0} = \sup_{t\in[0,T]}\|\mathbf{a}_t(0)\| < \infty$, the common Lipschitz bound implies

$$|q_{\mathrm{acc,t}}(x)| \le C_{a,0}\|x\| + \tfrac{1}{2}L_a\|x\|^2, \qquad \|\nabla q_{\mathrm{acc,t}}(x)\| \le C_{a,0} + L_a\|x\|.$$

The growth bounds place $\mathbf{a}_t$ in $\mathcal{G}_t$ by cutoff approximation. If the cutoff potentials are not smooth enough for the tests, convolution gives smooth compactly supported potentials whose gradients converge uniformly on a fixed compact set, hence in $L^2(p_t)$. Zero weighted divergence also makes $\mathbf{a}_t$ orthogonal to $\mathcal{G}_t$, so

$$\|\mathbf{a}_t\|^2_{L^2(p_t)} = 0.$$

At each fixed time, acceleration is zero almost everywhere. We need this conclusion along complete trajectories. Choose a countable dense set of times and intersect their full-probability sets of starting points. This intersection still has full probability. By continuity of $t \mapsto \mathbf{a}_t(X_t)$, acceleration is zero at every time along each selected trajectory. Hence $X_t = X_0 + tu_0(X_0)$ and $u_t(X_t) = u_0(X_0)$. Each Euler update from an exact point uses this constant velocity and reaches the next exact point. Euler is therefore exact on every finite increasing grid for the same set of starts. □

I Regular displacement interpolation

Regular displacement interpolation moves each source point to its target at constant velocity along a straight line. Euler follows these trajectories exactly. Under the assumptions below, the field also minimizes action. Tao & Choi (2026, Remark 7) note exactness on straight trajectories. The proposition gives sufficient assumptions on the map and proves exactness at successive grid points.

Proposition 15 (Euler exactness on regular displacement interpolation)**.** *Let $T > 0$ and let p_0 be absolutely continuous with a finite second moment. Let $\mathcal{Q} = \nabla\Phi_{\mathrm{OT}} : \mathbb{R}^d \to \mathbb{R}^d$, where $\Phi_{\mathrm{OT}} \in C^2$, and suppose*

$$0 < \ell_- I \preceq D\mathcal{Q}(x) \preceq \ell_+ I \qquad \text{for all } x$$

for fixed finite constants ℓ_- and ℓ_+. Define

$$\mathcal{T}_t(x) = (1-t/T)x + (t/T)\mathcal{Q}(x), \qquad p_t = (\mathcal{T}_t)_{\#}p_0,$$

and define the field by

$$u(t,\mathcal{T}_t(x)) = \frac{\mathcal{Q}(x) - x}{T}.$$

The maps $\mathcal{T}_t$ are global diffeomorphisms, which means that they are continuously differentiable bijections with continuously differentiable inverses. The field is jointly continuous, globally Lipschitz in space, and has linear growth. Its trajectories are $X_t = \mathcal{T}_t(X_0)$. On every finite increasing grid, Euler's method satisfies

$$\widehat{X}_k = \mathcal{T}_{t_k}(X_0)$$

for every initial point. It is therefore exact for particles and their distributions at the grid points.

The map $\mathcal{Q}$ is the quadratic-cost optimal map from p_0 to p_T. The displayed path is its displacement interpolation, and

$$\mathcal{A}_T(u) = \frac{W_2^2(p_0, p_T)}{T}.$$

The field also minimizes action among finite-action fields satisfying this fixed continuity equation, up to $dt\, p_t(dx)$-almost-everywhere equality.

Proof. The interpolating maps need global inverses to define the field at every position. With $s = t/T$, the symmetric derivative $D\mathcal{T}_t = (1-s)I + sD\mathcal{Q}$ has eigenvalues between $\min(1, \ell_-)$ and $\max(1, \ell_+)$. The lower bound gives strong monotonicity and injectivity, as well as properness, meaning that inverse images of compact sets are compact. The inverse function theorem makes the image open, while properness makes it closed. Since the image is nonempty and $\mathbb{R}^d$ is connected, the map is onto and has a global differentiable inverse. The uniform derivative bounds and field formula give the stated continuity, spatial Lipschitz continuity, and linear growth.

Each trajectory is affine in time. In particular,

$$\mathcal{T}_{t+h}(x) = \mathcal{T}_t(x) + h\frac{\mathcal{Q}(x) - x}{T}.$$

If Euler reaches the exact point at t_k, its next evaluation gives the same constant particle velocity and its update reaches $\mathcal{T}_{t_{k+1}}(x)$. Induction proves exactness on every grid.

By the convex-gradient characterization, $\mathcal{Q}$ is the quadratic-cost optimal map (Ambrosio et al., 2005, Section 6.2.3). Each particle has constant velocity, giving action $\mathbb{E}\|\mathcal{Q}(X_0) - X_0\|^2/T$. Since the map is optimal, this action equals the stated Wasserstein formula.

For minimum action with the path fixed, it remains to place the velocity in the weighted gradient closure. At $y = \mathcal{T}_t(x)$, the spatial derivative is

$$D_y u(t, y) = \frac{1}{T}(D\mathcal{Q}(x) - I)\{(1-s)I + sD\mathcal{Q}(x)\}^{-1}.$$

The matrices are symmetric and commute, so $D_y u$ is symmetric and u_t is a gradient. Linear field growth gives at most quadratic potential growth, allowing cutoff approximation under finite second moments. If a cutoff potential is only C^2, convolution gives smooth potentials with uniformly convergent gradients on a compact support. Thus velocity lies in the weighted gradient closure, and the projection identity proves minimum action on the fixed path. □

These map assumptions are sufficient for both exactness and action minimization. Euler exactness alone needs only a well-defined field at evaluation times along constant-velocity straight paths. For a singular optimal-transport map, the existence of such a field needs a separate argument.

J A scaled rotated Gaussian coupling

We scale the rotated Gaussian pairing of Liu (2022, Example 3.5) to obtain the same probability path as independent Gaussian interpolation, using dependent endpoints. In even dimension, let $\sigma > 0$, $J^\top = -J$, $J^\top J = I$, and $X_0 \sim \mathcal{N}(0, I)$. With $\mathcal{T}_t = (1-t)I + \sigma t J$ on $[0, 1]$, each particle follows a straight line to a scaled

rotation of its start. The endpoints are dependent, but $\mathcal{T}_t\mathcal{T}_t^\top = q_{\mathrm{G}}(t)I$ with $q_{\mathrm{G}}(t) = (1-t)^2 + \sigma^2 t^2 > 0$, giving the same Gaussian marginals as independent interpolation. The velocity field is

$$u(t,x) = (\sigma J - I)\mathcal{T}_t^{-1}x = \frac{\sigma^2 t - (1-t)}{q_{\mathrm{G}}(t)}x + \frac{\sigma}{q_{\mathrm{G}}(t)}Jx. \tag{51}$$

The radial term is the conditional-mean gradient velocity of independent Gaussian interpolation, while the rotation preserves each Gaussian marginal. Since $\mathcal{T}_{t+h} = \mathcal{T}_t + h(\sigma J - I)$, an Euler update from an exact point reaches the next exact point. Induction gives particle exactness on every grid, although rotation raises action above the gradient field by $d\sigma^2 \int_0^1 q_{\mathrm{G}}(t)^{-1}\,\mathrm{d}t > 0$.

For an anisotropic endpoint CX_0 with covariance $\Sigma_1 = CC^\top$, this deterministic linear construction matches the independent-interpolation path exactly when $C + C^\top = 0$. This follows by comparing the $t(1-t)$ covariance terms. An invertible real skew-symmetric matrix has two-dimensional blocks, so each eigenvalue of $CC^\top$ has even multiplicity. Conversely, equal eigenvalues can be paired into blocks $\sqrt{\mu}J$. The restriction applies only to this linear coupling class.

K Limits of action comparisons and evaluation schedules

Our examples use fields with positive additional action. In a larger class of fields, the same numerical improvement can require arbitrarily little additional action. Euler uses the field only at its evaluation times, while action integrates squared velocity over time. A continuous correction can keep the same values at evaluation times and vanish outside short intervals around them.

For the Gaussian probability path in Appendix A, fix a finite increasing grid and $a > 0$. At step j, set

$$b_j = h_j^{-1}\sqrt{e^{2ah_j} - (1+ah_j)^2}.$$

The inequality $e^{ah_j} > 1+ah_j$ makes b_j real and positive, with exact Euler covariance factor $(1+ah_j)^2+b_j^2h_j^2 = e^{2ah_j}$. Set $b(t_j) = b_j$ at each evaluation time and extend these values by continuous functions supported in disjoint intervals of width at most δ. Their sum b_δ is bounded and continuous, giving a global affine flow that preserves the Gaussian path. Shrinking the intervals keeps every Euler evaluation and output unchanged, while the added action satisfies

$$\operatorname{tr}(\Sigma_0)\int_0^T b_\delta(t)^2 e^{2at}\,\mathrm{d}t \longrightarrow 0 \quad \text{as } \delta \downarrow 0.$$

The grid is fixed and has finitely many evaluation times. The function heights are finite, so the action integral tends to zero as the total support length decreases. This gives the same numerical improvement with arbitrarily small additional action. The fields have no common bound on time derivatives.

L Numerical specification

We calculate distances between full probability distributions, without using sample estimates. For the Gaussian examples, recursive matrix updates check the distance formulas and particle maps. For the nonlinear example, optimal radial transport gives the distances. We use independent reference calculations to check the nonlinear distances and their leading coefficients.

L.1 Gaussian checks

We evaluate the scalar Gaussian formulas at 80 decimal digits. We check the particle maps, covariances, and means against recursive Euler matrix updates in binary64 double precision. Each update starts from the preceding numerical result, with the mean moving at constant velocity. To calculate the Gaussian Wasserstein distance, we optimally align the covariance square roots by an orthogonal map. This avoids subtracting nearly equal trace terms. Quadrature along the exact flow gives an independent check of continuous action.

The nonuniform grid is $(0, 0.04, 0.17, 0.43, 0.72, 1)$.

The Gaussian calculation uses $a = 0.6$, $T = 1$, and the fixed coefficients $b = 0$ and $b = a$. In dimension two, the covariance is

$$\Sigma_0 = \begin{pmatrix} 1.5 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}. \tag{52}$$

Uniform step counts are the powers of two from one through 4096. Dimensions four and eight use $1, 4, 32$ uniform steps. All three dimensions also use the nonuniform grid above. For each higher dimension d, set $\Sigma_0 = C_0 C_0^\top$, where

$$(C_0)_{ii} = 0.7 + 0.9\frac{i-1}{d-1}, \qquad (C_0)_{i,i-1} = 0.12 \ (i \geq 2),$$

and all other entries vanish. For every dimension, the mean and drift components are

$$(m_0)_i = -0.4 + \frac{i-1}{d-1}, \qquad \xi_i = 0.2 - 0.5\frac{i-1}{d-1}.$$

The matrix J uses consecutive two-dimensional quarter-turn blocks. We compare terminal squared distribution error, its maximum over grid points, and its sum over grid points. We also calculate error between particles from the same starting point and check the positive leading error coefficients in Appendix A.

The contraction control uses $a = b = -0.6$, the stated two-dimensional covariance, and one step on $[0, 1]$, with algebraic zeros recorded separately from floating-point residuals.

L.2 Nonlinear example and reference calculations

The nonlinear calculation uses $a = 0.6$, $\eta = 0.3$, $T = 1$, and source $0.6\mathcal{N}(0, 0.5^2 I_2) + 0.4\mathcal{N}(0, 1.5^2 I_2)$. The uniform step counts are powers of two from 8 through 4096. We also use the same unequal grid as above. For a spherical Gaussian component of scale s, $R_0^2/(2s^2)$ has density e^{-z} on $z \geq 0$. We therefore evaluate radius expectations by Gauss–Laguerre quadrature with 64, 128, or 256 nodes per component.

For the first reference calculation, we integrate log radius, angle, action, and the coefficient equations over time. We use SciPy 1.17.0's DOP853 ordinary differential equation solver, with primary relative and absolute tolerances 2×10^{-13} and 2×10^{-14}. For an independent reference, we separate variables in the radius equation. With $r > 0$, $a_+ = a + \eta$, and $\ell = \log(R(T, r)/r)$, this gives the implicit equation

$$\frac{\ell}{a} - \frac{\eta}{2aa_+}\log\frac{a + a_+ r^2 e^{2\ell}}{a + a_+ r^2} = T, \qquad aT \leq \ell \leq a_+ T.$$

This equation comes from integrating $1/f(r)$ between the source and terminal radii. Once we find the terminal radius, we integrate over radius to calculate angle, action, and leading coefficients. This calculation is independent of the time integration. Differentiating $\log f(R(s, r))$ gives $V(T, s; r) = f(R(T, r))/f(R(s, r))$ for $r > 0$, so

$$\mathcal{C}_i(T, r) = f(R(T, r)) \int_r^{R(T,r)} \frac{d_i(v)}{f(v)^2}\, \mathrm{d}v.$$

To separate quadrature error from reference error, we first increase the number of nodes with tolerances fixed. We then tighten the tolerances with the number of nodes fixed. The supplementary code includes all tolerances and comparison criteria.

We also compare independent particle updates in Cartesian and radius-angle coordinates, using source radii $(0.07, 0.4, 1, 3, 8)$ and angles $(0, 0.3, 1, 2.1, 4)$.

L.3 Solver, sensitivity, and interpolation checks

The solver calculations use $b = 0$, $b = a$, and $b = a/\sqrt{3}$ on the same grids. We check the scalar stability factors against separate matrix updates. To study Euler's sensitivity, we use fixed ratios $b/a \in \{0, 0.9, 0.98, 1, 1.02, 1.1\}$. The interpolation calculation uses $\sigma = 1.5$. Its corrected field has zero error algebraically, and we record floating-point residuals separately. These computations check the numerical evaluation of the formulas. The proofs give the convergence orders and strict comparisons.

L.4 Equal NFE

For the nonlinear equal-NFE comparison, we keep the same source and coefficients and use $N \in \{2^j : j = 1, \dots, 12\}$ evaluations. Euler takes N uniform steps, while base-field Heun takes $N/2$. We calculate distances using 128 Gauss–Laguerre nodes per component and compare them with results from 64 and 256 nodes. All methods use the implicit exact-radius reference. We check this reference independently by time integration at relative tolerance 2.5×10^{-14} and absolute tolerance 3×10^{-15}.

For base-field Heun, put $f(r) = \alpha(r)r$ and let $s = 2T/N$. One radius step is

$$H_s(r) = r + \tfrac{s}{2}\{f(r) + f(r + sf(r))\}.$$

Since $f' > 0$, its derivative is $1 + \frac{s}{2}\{f'(r) + f'(r + sf(r))[1 + sf'(r)]\} > 0$. The derivative is positive, so the radius map is increasing. The update also preserves angle. Pairing particles from the same starting point therefore matches angles and ordered radii. This pairing is optimal, so particle error equals W_2. We check all three methods with separate Cartesian updates. The smaller measured distribution error of corrected Euler applies to this source and these parameters.

Increasing quadrature from 128 to 256 nodes changes the errors by less than 1.8×10^{-7} relatively, while the independent reference discrepancy is below 2.6×10^{-8} of the corresponding numerical error.

The Gaussian Heun–RK3 comparison uses the parameters of Figure 2 at $12, 48, 192, 384$ field evaluations. We evaluate the complex polynomials at 50 and 80 decimal digits and check them against independent matrix-stage recurrences at 70 and 100 digits. The extra precision controls error from subtracting nearby values in the distance calculation.

L.5 Measured errors and refinement results

Table 4 gives the errors on the finest grid and order estimates $\log_2(e_{2048}/e_{4096})$, where e_N is error after N steps. We calculate the leading coefficients from independent radius integrals, without fitting the error curves. The primary calculation uses 128 nodes per mixture component. Increasing this to 256 changes every reported terminal error, action, and coefficient by less than 8×10^{-8} relatively. The largest reference discrepancy is below 3.2×10^{-7} of the corresponding Euler error.

Table 4: The nonlinear example has second-order corrected distribution error, although particle error remains first order. The calculation uses the source and parameters of Figure 5, 4,096 uniform steps, and 128 quadrature nodes per mixture component. Observed order $\log_2(e_{2048}/e_{4096})$ measures the error reduction when the step count doubles, where e_N is error after N steps. The leading coefficients are limits of error divided by h, h^2, and h in the three rows.

Quantity	Error	Observed order	Leading coefficient
Base distribution	3.391×10^{-4}	0.9997	1.3894
Corrected distribution	9.687×10^{-8}	1.9995	1.6258
Corrected particle	6.551×10^{-4}	0.9998	2.6835

The base and corrected actions are approximately 4.2342 and 8.6716.

The Gaussian sensitivity calculation uses up to 8,192 steps to resolve the first-order limit for a 2% coefficient mismatch. At this step count, all displayed scaled coefficients are within 0.5% of their asymptotic predictions. We check the smallest Heun errors with independent matrix recurrences at 50 and 80 digits, including values below binary64 resolution. At 4,096 steps, the error ratio between $b = a$ and $b = 0$ is 1.9998, consistent with the limiting ratio two.